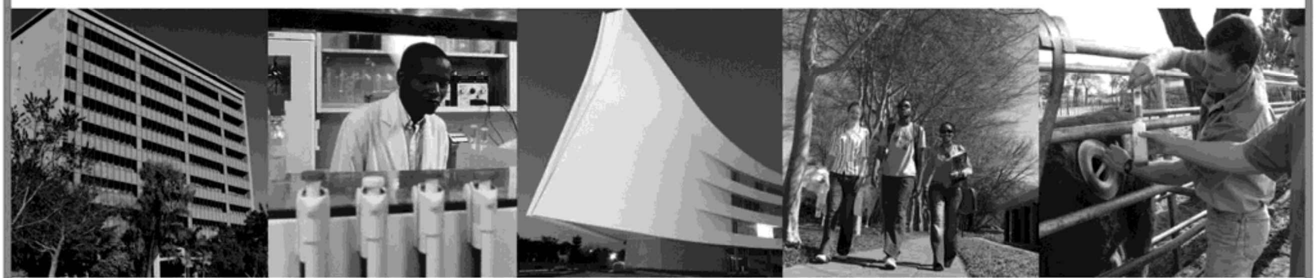
Fakulteit Natuur- & Landbouwetenskappe
Faculty of Natural & Agricultural Sciences

School of Physical Sciences

Department of Physics

LABORATORY MANUAL

FSK 116 & 176



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Experiments for engineering students

Contents

DEPARTMENT OF PHYSICS

The attendance of all practicum sessions is compulsory

- Failure to attend **all** sessions may result in examination refusal.
- Consult your study guide for rules and regulations.
- You can catch up the lab in the weeks allocated to catch-up labs. The dates for these can be found on clickUP. You may not simply attend any other session on another day to catch up when you have been absent.
- If you cannot attend a practical session for a valid reason (illness, driver's license appointments, car accident, etc) proceed as set out below to sign up for the catch-up lab session.

If you miss a practical / tutorial:
Within 5 days:

1) Download# both the Absent from Lab and Absent from Tutorial forms from the Practical folder on clickUP.
2) Complete the forms and attach copies of required documents to both forms.
3) Hand Absent from Tutorial form to your tutor during your tutorial session. Remember to check at a later stage whether he/she gave you "reason" and not zero for being absent.
4) Hand Absent from Lab form in at office 4-17 Natural Sciences Building1. Bring original documents(eg medical certificate)as proof of absence.

#Forms may also be collected from the office mentioned above.

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Dit is verpligtend om alle praktikumsessies by te woon

- Die versuim om **alle** praktikumsessies by te woon, kan tot eksamenweiering lei.
- Raadpleeg jou studiegids vir die reëls en regulasies.
- Jy kan die praktikumsessie inhaal in die weke daarvoor toegeken. The datums hiervoor kan op clickUP gevind word. Jy mag nie sommer uit eie inisiatief enige ander praktikumsessie op 'n ander dag bywoon om iets in te haal nie.
- Indien jy op grond van 'n geldige rede (siekte, bestuurlisensie-afspraak, motorongeluk, ens.) afwesig was, volg die instruksies soos hieronder uiteengesit noukeuring.

Indien jy afwesig was van die tutoriaal / eksperimentsessie:
Binne 5 dae:

1) Laai# die Afwesig-van-Lab en Afwesig-van-Tutoriaalvorms op clickUP af. Die vorms is in die "Practicals" vouer.
2) Voltooi die vorms en heg afskrifte van die nodige dokumente as bewys van verskoning van afwesigheid aan.
3) Gee die Afwesig-van-Tutoriaalvorm aan jou tutor gedurende die tutorsessie. Onthou om later te kontroleer dat hy/sy vir jou 'n "rede" en nie 'n nul toegeken het vir die tutoriaaltoets nie.
4) Handig die Afwesig-van-Labvorm by kantoor 4-17 Natuurwetenskappe gebou 1 in. Bring die oorspronklike verskoningsdokumente saam (bv siektesertifikaat).

#Vorms kan ook by bogenoemde kantoor afgehaal word.

Experiments for Physics Students

Why practical?

The basic aim of physics practical work is to help you to:

- Understand and apply the research process, in which data are gathered and processsed in order to derive patterns and relationships between variables.
- Experience physical phenomena firsthand.
- Gain an understanding of some of the physical concepts and theories of physics.
- Familiarise yourself with a variety of instruments and learn to make reliable measurements.
- Learn how a measurement can be made precisely with a given instrument and determine the measurement error.
- Do calculations so that results have the appropriate number of significant figures.
- Find functional relationships between data using graphical analysis.
- Write accurate and complete reports on experiments carried out.
- Learn that equipment sometimes does not function as expected, and how to recognise this when it occurs.

Assignments and Tests

In order to complete the experiment successfully and gain the full benefit from the practical it is essential that you prepare for your experiments **before** you come to the practical. After the third practical session you will write a short test on the theoretical background of the particular experiment or on the experiment itself at the beginning of the experiment.

You have to complete the experiment and **hand in** your report book before you leave the laboratory.

Practical Marks

The practical mark contributes between 10% and 25% to the semester mark. The final mark allocation for the year is given in your study guide.

For admission to the examination a 90% attendance of the practical and tutorials as well as a sub minimum of 50% in the practical is required.

All practical pre-tests and reports will be evaluated. At the end of the semester you might have to do a practical test. To pass this test, you must have mastered the study aims of each of the experiments.

The weighting of these marks is also given in your study guide.

Rules for the practical

- Experiments may **only** be done in the designated times. The work must be done in such a way that you can carry out the experiments or parts of the experiments without instruction.
- Your report book must stay in the laboratory and may not be taken home.
- It is a **requirement** that the experiments be **prepared beforehand** and that additional material, where necessary, be consulted.

- The right of admission to the laboratory and/or experiments is reserved.

- Bring your own ruler, calculator, pen and pencil to the practical.

- Please feel free to visit the Discovery Centre for its many interesting experiments and observations about the physical world.

- **All practical sessions are compulsory and only medical certificates or written proof of other circumstances will be accepted as a reason for absence from these sessions. You will have to make arrangements to work in any practical which you have missed during the catch-up labs.**

- Your demonstrator must sign your report book at every session.

- **It is your responsibility to see to it that you have done the experiment correctly and that you understand it.**

Guidelines for writing reports

A report has to be written on every experiment. This report should be written in the report book provided.

Pay attention to grammatical correctness and do not mix languages. Marks will be awarded for neatness and logical presentation of your data.

i. Heading

Use the three-letter code and description of the specific experiment as the heading.

ii. Objective

Briefly state the objectives of the experiment.

iii. Results

It is not necessary to describe the theory, experimental set up and method! Only your observations, results and conclusions should be reported in a systematic and organized way.

Give all your results and wherever possible, make use of tables and graphs, **even when not specifically requested to do so.** All graphs must be drawn according to the hints in your practical manual - see experiment 2.

Always take the **accuracy** with which your **measurements** were taken into account and round your answers off to the **correct number of significant figures!**

iv. Discussion and Conclusions

A brief discussion of your observations and results, even when not requested, is required. A few points to consider are:

- Are your results in agreement with the theory?

- What are the reasons for any observed deviation from the ideal result or behaviour?

- Can you think of experimental errors which could have influenced your results?

- Is it possible to estimate the effect of experimental errors on your observations and results?

- Do you have suggestions on how the results could have been improved? Do you have any suggestions or own theories?

Units

Every measurement of a physical quantity must be expressed in the applicable SI unit. The SI system is based on the units given in the table below. It then follows that although measurements in an experiment should be given in the units of the instrument, it is necessary to convert these units to **SI units** before they are used in calculations.

S.I. Units

The international system of units (Système International d'Unités) is based on the mksa (meter-kilogram-second-ampere) system. The fundamental quantities are:

Quantity	***Variable***	***Unit symbol***	***Unit***
length	ℓ	m	metre
mass	m	kg	kilogram
time	t	s	second
electric current	I	A	ampere
temperature	T	K	kelvin
luminous intensity	L	cd	candela
amount of substance	n	mol	mole

All other quantities, can be defined in terms of the fundamental quantities.

Quantity	***Variable***	***Unit Symbol***	***Unit***
area	A	m^2	square metre
volume	V	m^3	cubic metre
frequency	F	Hz	hertz
density	ρ	kg/m^3	kilogram per cubic metre
speed, velocity	v	m/s	metre per second
angular velocity	ω	rad/s	radian per second
acceleration	a	m/s^2	metre per second squared
angular acceleration	α	rad/s^2	radian per second squared
force	F	N	newton
pressure	p	Pa	pascal
work, energy, quantity of heat	W	J	joule
power	P	W	watt
electric charge	Q	C	coulomb
potential difference, electromotive force	V	V	volt
voltage	V	V	volt
electrical resistance	R	Ω	ohm
capacitance	C	F	farad
inductance	L	H	henry

Multiple and sub-units

Factor	10^9	10^6	10^3	10^{-3}	10^{-6}	10^{-9}
Prefix	giga	mega	kilo	milli	micro	nano
Symbol	G	M	k	m	μ	n

Errors

Every measurement contains an amount of uncertainty in it, that is, no measurement is 100% accurate. This is referred to as the error in a measurement. In the first place, the measuring instrument itself sets a limit on the how accurate a reading can be. A ruler for example, that is subdivided into centimeters, will measure the length of an object correctly only to the nearest centimeter. Its error will therefore be 0.5 cm (otherwise the measurement would have been estimated to the other centimeter mark). This type of error is called the **instrumental error**. Apart from instrumental errors we also have observational and personal errors that come into play - especially the errors of parallax and interpolation estimations. The determination of the error of a measurement is so important that one should supply both the measurement and an estimation of the error of that measurement, for example the diameter of a cylinder was measured as

$$d = 4.40 \pm 0.05 \text{ cm}$$

The 0.05 cm is called the **absolute error.** We could also write

$$d = 4.40\ (1 \pm 0.05/4.40) \text{ cm}$$

where 0.05/4.40 is called the **relative error** of the measurement.

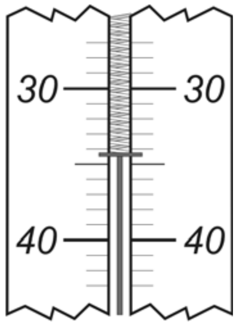
The **percentage error** is expressed as a percentage of the actual value:

$$\%error = \frac{measured\ value - actual\ value}{actual\ value} \times 100\%$$

There are methods to determine the error of a result. It is however, a difficult subject and for our purposes it is adequate to round the result off to the appropiate amount of **significant figures**.

Significant figures

Reading an analogue scale

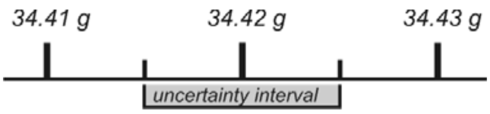


When taking a reading from an analogue scale you should estimate the least significant digit of the measurement by mentally subdividing the smallest division on the scale into more intervals. The diagram depicts part of a spring scale from which an object is suspended. The scale indicates the mass of the object in grams. One would report the measurement as **34.4 gram** in this case. Note that the final digit of the measurement (4/10 of a gram) was estimated and has an (not too unrealistic) amount of uncertainty. The suspended object does have a definite mass which we refer to as the **true mass** of the object. The measuring device attempts to determine this value. The **accuracy** of such a measurement is the amount by which the measurement differs from the true value. If the pointer of the spring scale came to rest exactly on the 34 gram mark on the scale one should report the mass as 34.0 gram. In such an instance the final digit would still have a certain measure of uncertainty. This reflects the situation that a measurement is only an approximation of the true value being measured. For analogue scales the measurement therefore should have one more digit than that given by the smallest division on the scale. Note that readings from a graph also entail reading an analogue scale.

Reading a digital scale

Suppose that we weigh the same object on an electronic scale with a **resolution** of 0.01 gram and the scale indicates its mass as in the diagram. Even in this case there is still a measure of uncertainty in the measurement since the **true mass**, m, of the object can be anywhere within the range

$$34.415 \quad < \quad m \quad < \quad 34.425$$



and the scale would still report the mass as 34.42 grams. The true mass of the object can be anywhere in the indicated interval according to the measurement. That is the mass can be reported as 34.420 ± 0.005 grams. In short it should be reported as 34.420 gram. So in this case we also add an additional significant digit to the measurement albeit a zero in all cases for a digital scale. The uncertainty will again be in the least significant digit of the measurement. This final zero is therefore one digit beyond the resolution of the instrument.

When are digits significant?

- All non-zero digits are significant, for example 27 N has two significant figures and 3.46 N has three significant figures.

When zeros appear in measurements one needs to consider their position in the value in order to decide whether they are significant.

- Leading zeros are not significant, they just serve to locate the decimal point. By using scientific notation one can ged rid of such zeros. For example 0.034 gram has only two significant figures.
- Zeros sandwiched between other digits are significant, so 20.03 ^{o}C has four significant figures.
- Zeros at the end of a measurement are considered not to be significant, that is 4 500 metres has only two significant figures. If the experimenter needs to indicate that they are actually significant then he/she should use scientific notation - see the next remark.
- Zeros after the decimal point are considered to be significant. That is 12.0 A has three significant figures and $4.500 \times 10^{3}\ m$ has four significant figures.

Significant figures and measurements

In the example above we see that measuring the object with the spring scale gave us a mass of 34.4 grams while the electronic scale gave us a mass of 34.420 grams. The price of the spring scale is about R50 and that of the electronic scale R5000. So the extra two digits in the second measurement came at a substantial price. In the experimental world it usually takes much more than just money to get a more accurate value for a certain measurement. It usually involves more time, effort and ingenuity to get those extra digits. **Take care in reporting measurements to the correct number of significant figures.**

Calculations with measurements

Let's assume that we are actually interested in the weight, *W*, of the object. It can be calculated using the value $9.8\,m/s^2$ for gravitational acceleration. Since the second measurement came at a higher price one would expect that the weight calculated from its measurement should have more significant figures to it. That is for the spring scale we would expect that the weight should be reported as 0.337 N and for the electronic scale 0.33732 N, while keeping the answers to the same number of significant figures as the measurements. In actuality the gravitational acceleration is also a measurement with its own uncertainty. It follows logically that the uncertainty in its value will also have an influence on the number of digits in the calculated weight of the object. So how does one decide how many significant figures there should be in a value calculated using measurements?

Significant figures in calculated values

We use some simple rules to decide how many significant figures the final value should be rounded off to. In **calculations involving multiplication, division, trigonometric and other functions** we round the final value off to the the same number of significant figures as the measurement with the least number of significant figures. The weight of the object would therefore be reported as 0.34 N in both cases due to the fact that the gravitational acceleration has only two significant figures. The situation can be improved upon if we can obtain a value for *g* with more significant figures (more money, time and effort). As a further example if we were to calculate the value of $\sin(kx)$ using the values $k = 0.097\,m^{-1}$ and $x = 4.73\,m$ we obtain the answer 0.44 since *k* has only two significant figures.

When the value being calculated is obtained from **additions and/or subtractions** of values the final value is rounded off to the same **decimal** as the measurement with the least number of decimals, for example:

$$\begin{array}{ll}
5.67\,J & 2\,decimal\ places \\
1.1\,J & 1\,decimal\ place \\
\underline{0.9378\,J} & 4\,decimal\ places \\
7.7\,J & 1\,decimal\ place
\end{array}$$

Exact values

Numerical constants have no uncertainty in their value.They are therefore not measurements and do not influence the uncertainty in a value calculated from them. Some examples are the number of millimeters in an inch (2.54 mm), which is a defined quantity (the inch) or π, which can be calculated to any required precision or the number 2 (a counting value) in the formula $2\pi r$. Such values are not taken into consideration when deciding how many significant figures there should be in the final answer.

Intermediate calculations

When a calculated value is the result of several steps of calculations one should keep (at least) one more significant digit in the intermediate answers than that which is required in the final value.

Scientific notation

Very large or small values are best reported using scientific notation. In this notation the decimal point is repositioned just after the first significant digit in the value. For example to report 14297 J in scientific notation the decimal point needs to be positioned just after the digit 1. That is we need to divide the value by ten thousand or multiply it by 10^{-4}. In order not to change the value we therefore also need to multiply it by ten thousand or 10^4:

$14297\,J$

$$= 14297 \times 10^{-4} \times 10^{+4}\ J$$

$$= 1.4297 \times 10^{4}\ J$$

Graph Rules

Rules for drawing graphs

1. The graph should have a concise title. All symbols that are used in the graph must be mentioned in words in the heading and the subject or apparatus of the experiment should be mentioned, e.g. The temperature change, ΔT, of a sample of water as heat, Q_W , is added to it.

2. The independent variable (the quantity that is changed **directly** by the experimenter) is plotted on the horizontal axis (the abscissa) and the dependent variable (the quantity that changes in the experiment as a result of changing the independent variable) is plotted on the vertical axis (the ordinate).

3. Clearly indicate along each axis the variable plotted on the axis in words as well as the relevant units e.g. resitance(Ω).

4. The scales of the two axes, not necessarily the same, should be chosen so as to allow the graph to occupy most of the available space. The axes do not need to start at zero.

5. The intervals should be chosen to **facilitate** plotting and reading of points.

6. Mark each experimental point with a dot with a circle around it.

7. If the points lie more or less on a straight line use a ruler to draw a straight line in such a position that the points are evenly spaced about it. This is called the line of best fit. Otherwise draw a smooth curve through the points. This reduces the effect of experimental errors. Never connect the points by means of short straight lines.

8. **The gradient of a graph**

   The gradient (or slope) of a graph is **always** calculated by means of the definition:

   $$gradient = \frac{y_2 - y_1}{x_2 - x_1}$$

   This definition is also used when the graph passes through the origin. **Never** calculate the slope using m = y/x.

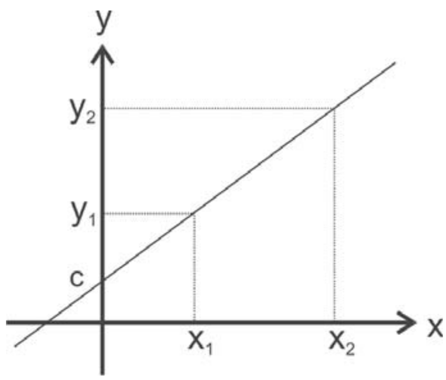


Fig.1 a Linear relationship.

- Always determine the slope from the fitted line, **never from the experimental points.**

- Choose the largest possible Δx and Δy, this allows for more accurate results. In other words, choose two points that are as far apart as possible.

- If the slope of any region of a graph is not constant, it is called a curve, as in **figure 2**. The slope of a curve at any given point is equal to the slope of the tangent at that point. The slope of the tangent is calculated in the same manner as for a straight line. The gradient of a curve is therefore not constant and depends on the independent variable.

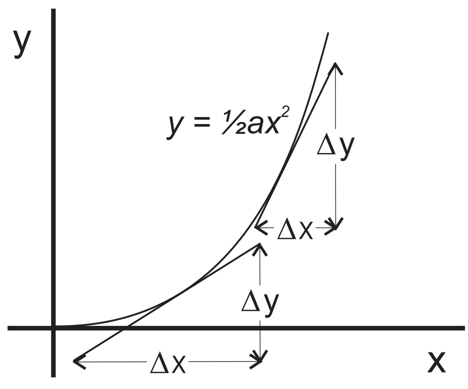
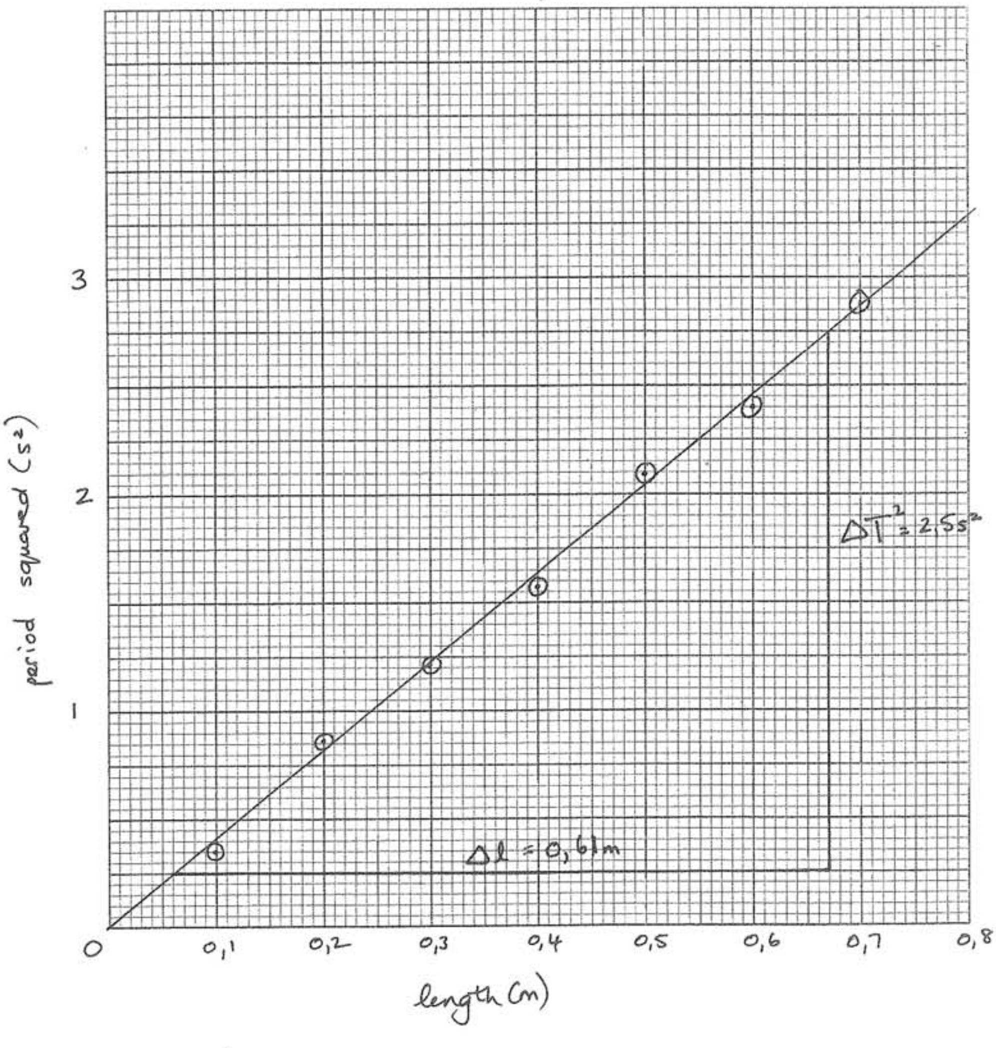


Fig. 2 The gradient of a curve.

9. If the relationship between the dependent and the independent variable is linear as in **figure 1**, then the relationship between the two quantities is of the form: $y = mx + c$

The square of the period of a simple pendulum as a function of its length.



where m is the gradient of the graph and c is the y-intercept. This equation with the aforementioned constants is called the empirical (obtained through experiment procedures) equation. By comparing the empirical equation with its theoretical counterpart it is possible to determine some useful constants.

The empirical equation of the graph in the example on page 10 is

$$T^2 = 4{,}1(s^2/m)\,\ell + 0$$

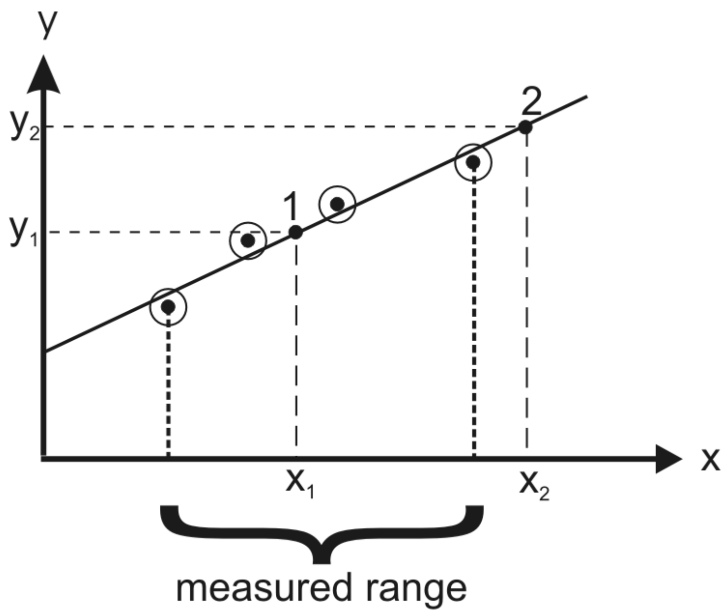


Fig.3 Interpolation and extrapolation.

10. A graph can be used to read off values for the dependent variable at values of the independent variable at which no measurement were made, like at x_1 and x_2 in **figure 3**.

    If the reading is made at a value for the independent variable that is within the experimental measured range then the process is called **interpolation**, as at point 1. Interpolation leads to relatively reliable values for the dependent variable.

    If the reading is made at an independent value outside of the measured range then the process is called **extrapolation**, as at point 2. Such values are not as reliable as those obtained by interpolation since the functional relationship may deviate from a linear relationship outside the measured range. Rather extend the measuring range to include the value at which a reading is required if this is at all possible.

References:

1. Fishbane: Physics for Scientists and Engineers
2. Giancoli: Physics
3. Halliday: Fundamentals of Physics
4. Young et al: University Physics
5. Halliday et al: Fundamentals of Physics

GRA 1

Graphs

Introduction

The quantitative information collected during an experiment is known as data. Data is used to determine a functional relationship between the variables of an experiment. This functional relationship can be represented in three ways:

a. a table of numbers
b. an equation
c. a graphical representation

The following section describes the important rules and techniques for representing data graphically.

The use of graphs

1. A graph is a visual representation of the functional relationship between two variables.

2. Graphs may enable one to determine the mathematical relationship between the variables. This mathematical relationship is called the empirical relationship. If the theoretical relationship between the variables is also known then some relevant constants can be determined by comparing the theoretical and empirical equations - for example **graph D** on page 19.

3. Graphs show systematic deviations of the data from the general trend of the graph.

4. Graphs make it possible to see where small changes in the independent variable cause large changes in the dependent variable.

5. Interpolations and extrapolations can easily be made on linear graphs - for example **graph A** on page 14.

Rules for drawing graphs

The rules for the drawing of graphs are summarised in the introductory section of this manual. These rules should always be observed when constructing graphs.

1. The straight-line graph y=mx +c

If a variable y is directly proportional to another variable x, i.e. $y \propto x$ or $y = mx$, then the graph will be a straight line which passes through the origin (**figure 1**). **m** , the proportionality constant, is the gradient of the graph.

The gradient or slope of the graph is calculated with rise over run:

$$m = \frac{\Delta y}{\Delta x}$$
$$= \frac{y_2 - y_1}{x_2 - x_1}$$

Remember to take the scales and units of the axes into account when calculating the gradient. The units of the gradient are the units of the y-variable divided by the units of the x-variable.

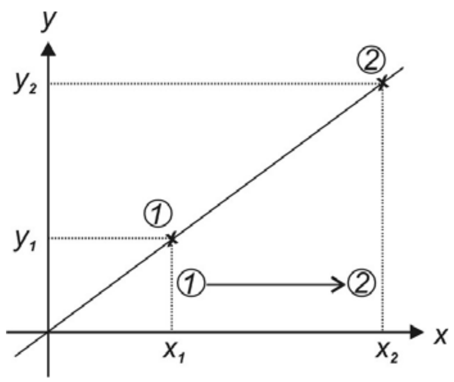


Fig. 1 A directly proportional relationship.

If the change in y is directly proportional to the change in x, $\Delta y \propto \Delta x$ (**figure 2**) we have that the graph will be a straight line that does not pass through the origin. Such a graph can be represented by the equation:

$$y = mx + c$$

where c is the intercept on the vertical (y) axis. See **figure 2.**

$$m = \frac{\Delta y}{\Delta x}$$

$$= \frac{y_2 - y_1}{x_2 - x_1}$$

The gradient m is, calculated as before

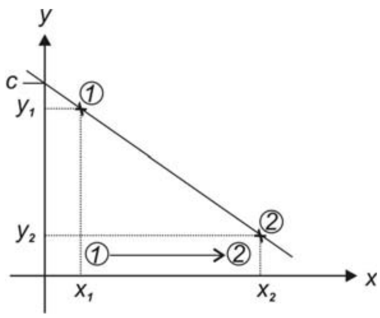


Fig. 2 $y = mx + c$ graph with a negative gradient

Graph A

The resistance r of a homogeneous wire has been measured at different temperatures t. The results are shown in **Table 1**.

Table 1 The resistance of a homogeneous wire at different temparatures.

Temperature t(°C)	*Resistance* r(Ω)
11,4	158
22,2	162
30,1	168
41,3	176
48,7	181
60,6	184
68,7	193

1. Draw a graph of the resistance as a function of the temperature of the wire. The origin of the temperature axis should be zero, but for the sake of convenience, the origin of the resistance scale should not be zero.

   Now that we have drawn a graph of the data and found that there is a linear relationship between the variables we can formulate the mathematical relationship between the involved quantities. This formula is called the empirical relationship since it was determined with experimental data.

2. Determine and clearly indicate on your graph, the change in resistance per unit change in the temperature, that is $\Delta r/\Delta t$, or gradient of the graph. Choose points 1 and 2 as far as possible from each other - see **figure 1** - and do not use data points for them. Do not forget the units of the gradient when you state its value.

3. Set the empirical equation up for the resistance of the wire at different temperatures. That is the formula relating the relationship between the resistance and temperature of the wire. This equation should have values and units for the constants in the equation. Use the symbols r and t for the variables in the equation.

4. Now that we have determined the relationship between the measured quantities we can either use the graph or the empirical equation to determine values for the resistance of the wire at other temperatures. Let's use the graph to determine the resistance of the wire at the following temperatures. Also indicate on it where the readings were taken:

   i. The resistance of the wire at 5°C. We are extrapolating our data in this case. That is assuming that the functional relationship will remain linear. Such an assumption is risky since we do not know whether the relationship is linear outside of the measured range of values.

   ii. The resistance at 25°C. Now we are interpolating our data, a much safer process.

2. The power function $y = kx^n$

The power function is often found in physics, for example:

a. The distance y that an object, initially at rest, falls under the influence of gravity during a time interval t is given by:

$$y = \tfrac{1}{2} g t^2$$

b. The period T of a simple pendulum of length L is $T = k\sqrt{L}$.

c. The force F between two point charges separated by a distance r is given by an inverse square law $F = \frac{k}{r^2}$

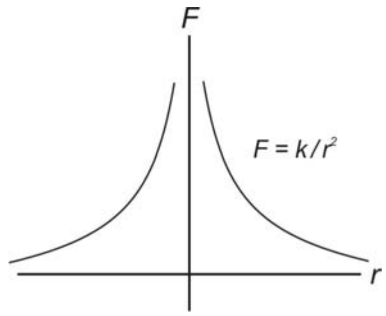


Fig.3 The force between two point charges.

Advantages of linear graphs

a. It is easy to fit a straight line to the data points.

b. The formula which relates the relationship between the variables can be established.

c. Interpolation and extrapolation - determining values for the dependant variable other than in the measurement set - can easily be made on a straight line graph.

It is therefore desirable to transform a graph into a straight line, which will be discussed next for one type of relationship between the variables.

Transforming a power relationships to a straight line graph

If there is a power relationship between two measured quantities

$$u = kv^n \quad \text{..................}\mathbf{(1)}$$

the resultant graph will not be linear. In such a case we find that plotting the logarithms of the measurements will conform to a straight line - see **graph B** for an example.

If we take the logarithm on both sides of the equation above

$$log(u) = log(kv^n)$$

$$= \log(k) + \log(v^n)$$

$$\underset{\substack{\uparrow \\ y}}{log(u)} = \underset{\substack{\uparrow \\ c}}{log(k)} + \underset{\substack{\uparrow \\ m}}{n}\ \underset{\substack{\uparrow \\ x}}{log(v)}$$

we obtain a linear relationship between the logarithms of the dependent and independent values, that is we get a linear graph from the "new" set of data values, $x = \log(v)$ and $y = \log(u)$. From such a linear graph we can then determine the coefficient, k, and the exponent, n, of the original power relationship . Note that the y-intercept and gradient of the linear graph will be such that $c = log(k)$ and $m = n$.

If the power relationship between the original measured quantities is known, or assumed, then a graph of u against v^n will result in a straight line graph with slope k for the "correct" value of n - for example **graph D**.

Logarithmic graph paper

A log-log graph can be drawn by first determining the logarithms of the numbers and then plotting the logarithms on normal linear graph paper. To save the effort of calculating the logarithms, special graph paper exists that was prepared by drawing grid lines at logarithmic intervals. Such log-log graph paper, where both axes are divided according to a logarithmic scale, will be used to plot the following graph.

Graph B - Fitting a power function to data

We will investigate the relationship between the semi-major axes of the orbits of the planets, a, around the sun and the time, T, to complete their orbits. For your interest, below is a short summary of the history of this investigation.

Kepler's analysis of Tycho Brahe's planetary data convinced him that the motion of the planets can be described mathematically. Both his first and second laws strongly suggested that the sun plays a central role in their motion. He needed evidence that the sun was actually responsible for their motion. This came from the fact that the orbital speeds of the outer planets are less than those of the inner planets. On this topic he wrote to one of his colleagues in a letter as follows "*...we must choose between two assump- tions: either the souls which move the planets are the less active the farther the planet is removed from the sun, or there is only one moving soul in the center of all the orbits, that is the sun, which drives the planet the more vigorously the closer the planet is, but whose force is quasi-exhausted when acting on the outer planets because of the long distance and the weakening of the force which it entails*." and further "*My aim is to show that the heavenly machine is not a kind of divine, live being, but a kind of clockwork, insofar as nearly all the manifold motions are caused by a most simple, magnetic, and material force, just as all motions of the clock are caused by a simple weight. And I also show how these physical causes are to be given numerical and geometrical expression*."[1]

***Table* 2 The semi-major axes and periods of orbit of some solar planets.**

Planet	a (a.u.)	T (yr)
Mercury	0,386	0,241
Venus	0,721	0,615
Earth	1,00	1,00
Mars	1,52	1,88
Jupiter	5,19	11,9
Saturn	9,53	29,5
Uranus	19,1	84,0

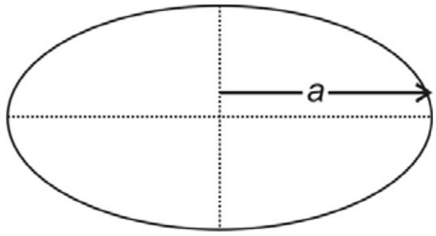


Fig. 4 The semi-major axis of an ellipse.

Let's use the data of the planets in **Table 2** to provide some evidence for his view. The values *a* represents the semi-major axes of the orbits of the planets, in astronomical units (the average sun-earth distance), and *T* the time to complete their orbits about the sun.

We intend to try and fit a power function of the form

$$a = kT^n$$
$$\therefore \log(a) = \log(k) + n\log(T) \quad \textbf{..................(2)}$$

to the data.

1. Plot the data on the log-log graph paper at the back of your report book.

   We can conclude from the fact that the graph seems to be linear that we can fit a power function to the data. Note that a graph of $\log(a)$ *vs* $\log(T)$ drawn on normal graph paper will also be linear.

2. Determine the gradient of the graph according to

   $$gradient = n$$
   $$= \frac{\log(a_2) - \log(a_1)}{\log(T_2) - \log(T_1)}$$
   $$= \frac{\log(a_2/a_1)}{\log(T_2/T_1)}$$

3. Read the intercept off on the *a* axis of the graph where the period has the value $T = 1.0$ *yr*. This is the constant *k* in the equation above. Why don't we take the log in this case? The units can be determined by considering the units of the variables, *a and T*, in **equation (2)**.

4. Next set up the empirical relationship between the distance and period values. Supply values for the constants, together with their units, in the equation.

5. Does this analysis lend weight to Kepler's views on the nature of the motion of the planets? Explain.

The relationship between the average distances of the planets from the sun, or their semi-major axes, and their times to orbit it is also known as Kepler's harmonic ("fitting together") law.

1 Arthur Koestler, *The Sleepwalkers*, Arkana Books, London, 1989.

GRA 2

The exponential function $y = a^x$

Exponential functions are described by a mathematical relationship of the form $y = a^x$. Note that the variable is in the exponent. a is the base and it is raised to the power of the controlling variable, x. We use only positive values for the base.

We are dealing with an exponential function whenever the growth rate of a certain quantity is constant. For instance the growth rate of the human population is approximately 1,53% per year. That means that if the population was p_1 in a particular year it would be $1,0153\,p_1$ in the following year. The startling nature of this type of function is that although the growth rate might seem small the quantity itself increases (or decreases) quite fast (exponentially) over time. Examples of such processes are:

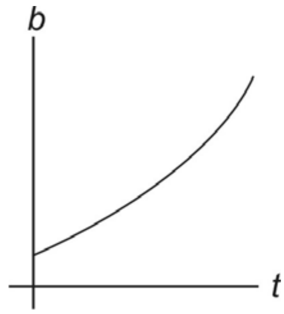


Fig. 1 The amount of bacteria in a colony with time.

a. The decay of radioactive material.The amount of radioactive isotopes decreases exponentially with time as they mutate to other elements. In this case the relation is of the form $y = a^{-x}$ to reflect that the quantity is decreasing.

b. The growth rate of a colony of bacteria, given enough resources, is directly proportional to the size of the colony at that particular time. That is, the amount by which they change per unit time, or db/dt, becomes more as the amount of bacteria that is present increases, or $db/dt \propto b$. This is evident in the fact that the gradient of the graph of the amount of bacteria present as a function of time increases as the amount of bacteria increases - see **figure 1**. The proportionality constant is the growth rate, r, of the colony (which is constant). We incorporate the growth rate as follows into the exponential function

$$b = a^{rt}$$

In this way it controls how fast the function is increasing.

c. The emptying of water from a tank.

d. The reduction in air pressure with height.

e. The discharging of a capacitor in a circuit. The rate at which the capacitor discharges decreases as the amount of electric charge on it decreases, similar to the rate at which a colony of bacteria changes.

For naturally occuring processes we often find that the base $e = 2,\dot{7}182\dot{8}$ should be used. The exponential function then becomes

$$y = y_o\, e^{-rx}$$

for decreasing functions and

$$y = y_o\, e^{rx}$$

for increasing functions. y_o is the starting value for the function, that is when the control variable, x, is zero.

Logarithms

If $y = a^x$ then we say that the logarithm of y to the base a is x or $\log_a(y) = x$. The logarithm of a number is therefore the exponent to which the base must be raised in order to obtain that number.

Common logarithms are to the base 10. In such a case the base is not indicated explicitly, for example $\log(100) = 2$. We speak of natural or Napierian logarithms when the base is e. In such a case we write ℓn rather than $\log_e$, for example $\ell n(2,00) = 0,693$.

Transforming graphs of exponential functions to straight-line graphs.

If one physical quantity, y, is decreasing exponentially with respect to another, x, then

$$y = y_o e^{-\lambda x}$$

Taking the natural logarithms on both sides of this relationship we get

$$\ell n(y) = \ell n(y_o) + \ell n(e^{-\lambda x})$$

$$\ell n(y) = \ell n(y_o) + (-\lambda) x \quad \text{...........} \textbf{(1)}$$

$$\uparrow \qquad \uparrow \qquad \uparrow \; \uparrow$$

$$y \; = \; c \; + m \; x$$

We can therefore transform an exponential graph to a linear graph if we plot a graph of $\ell n(y)$ as a function of x. Note the the gradient of the linear graph will be the negative of the decay constant, λ, that is $m = -\lambda$, and its y-intercept will be the natural logarithm of the starting value or $c = \ell n(y_o)$.

Semi-logarithmic graph paper

To save the effort of first calculating the logarithms special graph paper called semi-logarithmic graph paper has been designed with the grid lines drawn at logarithmic intervals along one axis and linear divisions along the other. Such graph paper is used to draw exponential functions.

Graph C - Fitting an exponential function to data

Quantities that increase or decrease exponentially are relatively common. The question is, how would one recognise it from a set of data and how would one set up the mathematical relationship between the variables? In this section we intend to answer these questions.

The count rate of a radioactive sample was measured with a radiation detector over a period of 20 minutes. The data is displayed in **table 1**. We intend to fit an exponential function of the form

$$R = R_o e^{-\lambda t} \quad \text{..................} \textbf{(2)}$$

to the data.

Table 1 The decreasing activity of a radioactive sample.

time(minutes)	count rate (counts/s)
0,0	800
5,0	400
10,0	205
15,0	105
20,0	50

1. Draw a graph of the count rate, R, of the sample as a function of time. Try and draw a smooth curve through the data points. Note that the relationship is not linear.

2. Make a new table with x-values the time in minutes and the y-values the natural logarithm of the count rate values.

3. Draw a graph of this new set of values. It should be linear if the original data can be represented by an exponential function. Note that one would also obtain a linear function if the data were to be plotted on semi-logarithmic graph paper. The fact that these graphs are linear confirms our suspicion that one can fit an exponential function to the data. You will use semi-logarithmic paper in the EKS experiment later.

4. Set up the empirical relationship between the x and y quantities in the graph. Determine the y-intercept, y_o, and the slope, α, of the graph. Choose points 1 and 2 as far apart as possible on the graph and do not use data points. Also do not forget to indicate the units of the constants in the equation!

5. Rewrite the empirical relationship using the real origin of the variables - that is $x = t$ and $y = \ell n(R)$. You should now have a relationship of the form

$$\ell n(R) = y_o + \alpha t$$

6. Note that if we take the natural logarithms on both sides of **equation (2)** we obtain

$$\ell n(R) = \ell n(R_o) + (-\lambda)t$$

We can now see from the last two equations that we can transform the empirical relationship into an exponential relationship if we set $y_o = \ell n(R_o)$ and $\alpha = -\lambda$. Calculate corresponding values for R_o and λ from the slope and y-intercept (units!).

7. Set up **equation (2)** with the values (and units) for the constants. This is our exponential function for the data.

8. Using **equation (2)**, derive the equation given below, where $T_{\frac{1}{2}}$, the half-life, is the amount of time taken for the activity to be reduced to half its value.

$$\ell n(2) = \lambda T_{½}$$

9. Calculate the half-life, $T_{\frac{1}{2}}$, with this relationship using the value of the decay constant as found in **point 6**.

Summary

For an exponential relationship between the variables we obtain a linear graph on normal graph paper if we plot the natural logarithm of the dependent values versus the independent values. An exponential function also produce a straight-line graph if it is drawn on semi-logarithmic graph paper.

Power functions are linear if drawn on log-log graph paper. On normal graph paper they produce a linear graph if one plots the common logarithm of the dependent values versus the common logarithm of the independent values.

Given a set of measurements of two quantities, how would one determine the empirical relationship between them by drawing a graph?

- First plot the data on axes with linear divisions. One can readily set up the empirical relationship if such a graph is linear. It is a direct proportion.

- If a straight line is not obtained, plot one set of values against the inverse of the other. If this is a straight line, we have an inverse proportion.

- If a straight line is still not obtained, we resort to graphs in which the divisions on one or both axes are logarithmic.

  - First try a semi-log graph: One axis has linear divisions and the other axes has logarithmically spaced divisions. A straight line suggests that one can fit an exponential relationship to the data.

  - If this is unsuccessful one can plot the data on a double - log plot, where both axes have divisions that are logarithmically interspaced. A straight-line graph then suggests a power relationship can be fitted to the data.

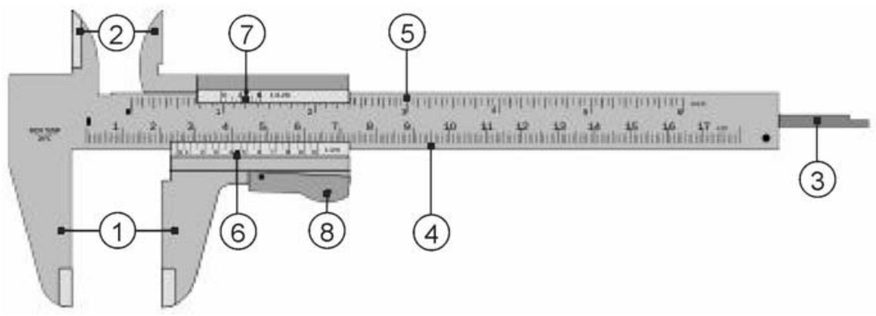
MET

Measurement and units

Introduction

We often need to measure quantities in physics experiments. In this experiment you will use two types of measuring devices - a vernier caliper and an electronic scale.

A vernier caliper is used to accurately measure the linear distance between two solid surfaces very precisely. Here you need to learn how to use and read it off correctly.



An example of a venier caliper

1 Outside jaws - used to measure the external diameter or width of an object
2 Inside jaws - used to measure the internal diameter of an object
3 Depth probe - used to measure the depth of an object or hole
4 Main scale - gives measurements up to one decimal place (in cm)
5 Main scale - gives measurements in inches. Note: Some calipers might not have these.
6 Vernier - gives measurements up to three decimal places (in cm)
7 Vernier - gives measurements in fractions of an inch
8 Retainer - used to lock the position of the movable jaws and probe. Note: Other types might have a locking screw on top.

You will also measure the mass of an object on a digital electronic scale. You need to check that the scale is leveled and zeroed when using it. Some scales can be calibrated and they actually measure the mass and not the weight of an object (which would change due to the variations in the gravitational acceleration over the surface of the earth!). A scale with a large scale pan might also be sensitive to drafts due to the Bernoulli effect.

Study Aims

You must be able to:

1. Use the vernier caliper with the necessary dexterity and accuracy to make measurements.

2. Use an electronic scale to determine the mass of an object.

3. Convert measurements to S.I. units.

4. Round calculated values off to the correct number of significant figures.

A. Significant figures

Indicate the number of significant figures in the following values:

a. $4{,}3 \times 10^3$ km

b. 0,0160 s

c. 200 g

d. What is the difference between the following two measurements: 2 cm and 2,0 cm?

Use the densities in **Table 1** for the following calculations. The densities in the table are also measurements so their significant figures need to be taken into account! Work in S.I. units and round your final answers off to the correct amount of significant figures.

e. Calculate the mass of a bakelite plate with dimensions 35,0 cm x 57 mm x 1,000 m.

f. Calculate the mass of a spherical perspex bead with a diameter of 13,0 mm.

Digtheid / *Density* ($\times 10^3$ kg/m³)	**Substance**
0,00120	air
0,3 tot/to 0,9	wood
0,92	ice
0,880 tot/to 0,940	polyethylene
1,00	water
1,06	blood
1,18	perpex
1,33	bakelite
1,3 tot/to 1,45	PVC
1,7 tot/to 2,0	bone
2,20	teflon
2,70	aluminium
2,5 tot/to 4,2	glass
7,87	iron
7,75 tot/to 8,1	stainless steel
8,40 tot/to 8,73	brass
8,96	copper
10,5	silver
11,34	lead
13,6	mercury
19,1	uranium

Table 1 Densities of some substances.

B. Vernier caliper readings

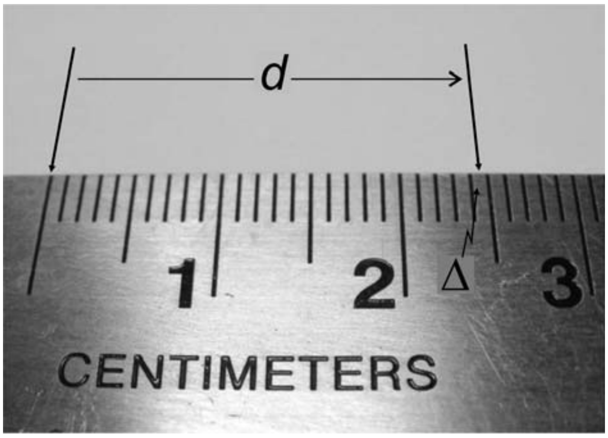


Fig. 1 A centimeter ruler reading.

1. What is the length, *d*, in millimeters in **figure 1** ? Include an estimate for the fraction of the millimeter - Δ - in the value for *d*.

2. The jaws of the caliper are closed in **figure 2** and open in **figure 3**. How many millimeters are they open in **figure 3**? Estimate the fraction of the millimeter - Δ - again.

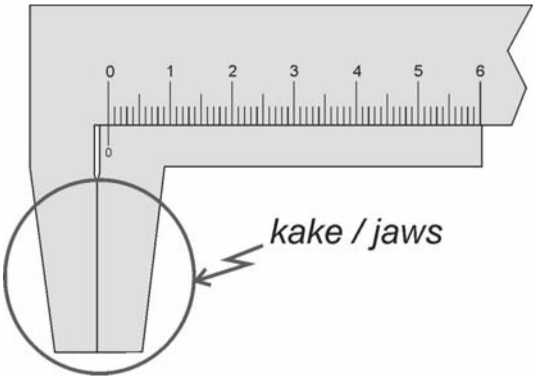


Fig. 2 A vernier caliper with its jaws closed.

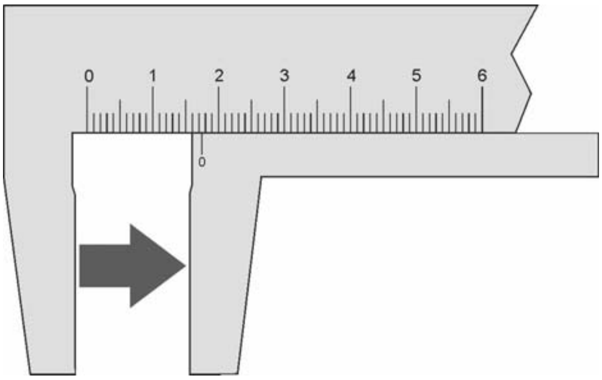


Fig. 3 A vernier caliper with its jaws open.

3. **Figure 4** shows the vernier scale on the moveable jaw of a vernier caliper. The 0,02 mm indicates that this scale enables one to take readings with a 0,02 mm resolution. We say that the least count of the caliper is 0,02 mm. The line to the right of the 0 line is the 0,02 mm line. To its right is the 0,04 mm line, then the 0,06 mm followed by the 0,08 and the 0,10 mm with a 1 digit next to it. Note the final 0 in 0,10 mm! What vernier line is the arrow pointing to on the scale?

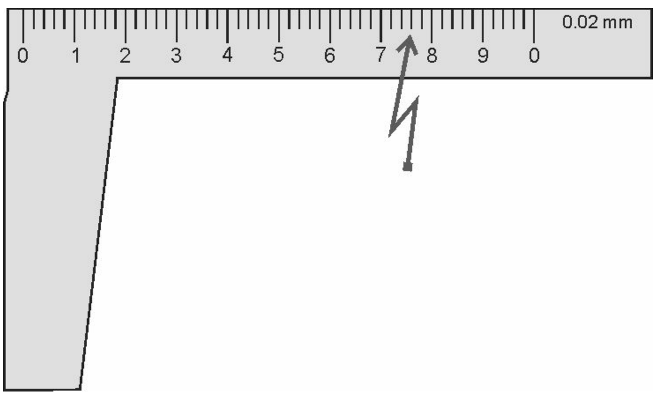


Fig. 4 A 0.02 mm vernier scale.

4. The vernier scale enables one to accurately determine the value of Δ, the amount by which the zero vernier line went over the millimeter mark, or the fraction of the millimeter value of the reading. The vernier reading is made at the point where a vernier line aligns with a millimeter line on the centimeter scale above it. Alignment occurs at the arrow in **figure 5**. What is the vernier caliper reading in millimeters including the fraction of the millimeter or vernier reading?

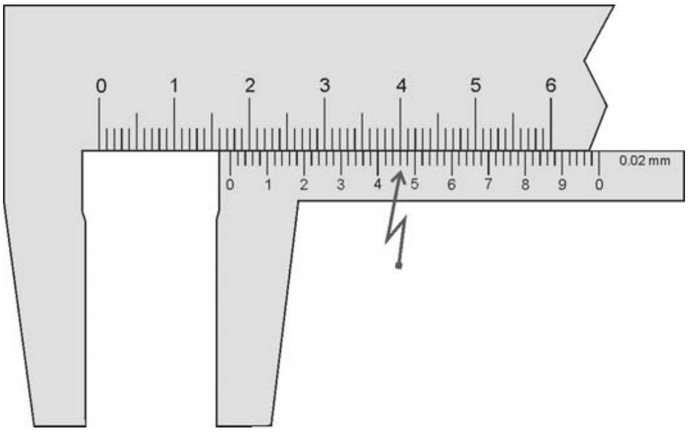


Fig. 5 A reading on a vernier caliper.

5. What is the vernier caliper reading in **figure 6**?

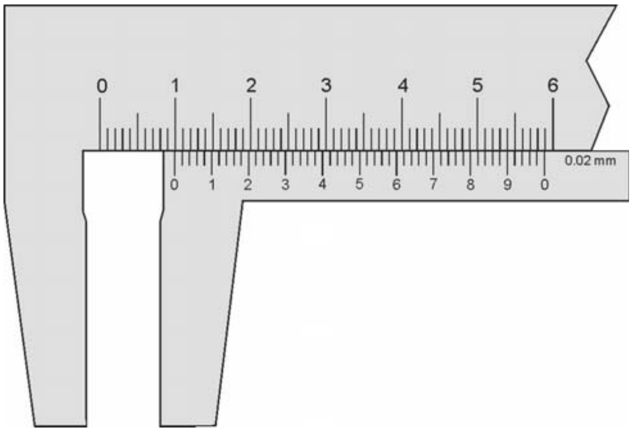


Fig. 6 Another reading on a vernier caliper.

6. The least count of the vernier caliper in **figure 7** is 0,05 mm. What is reading in this case?

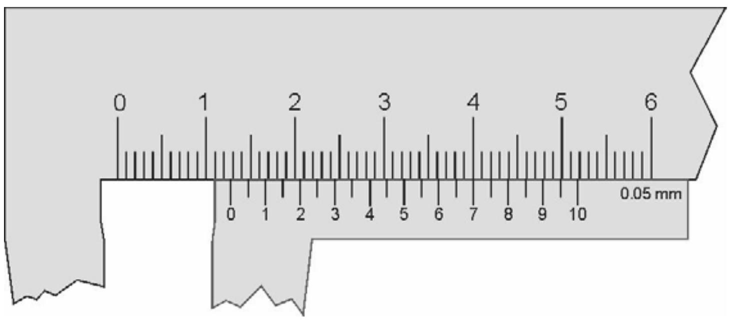


Fig. 7 Another reading on a vernier caliper.

C. Density determination

Next we want to determine the density, ρ, of the substance of a piece of metal square tubing. The density can be calculated using:

$$\rho = \frac{m}{V}$$

where m is the mass of the tube and V the volume of the metal.

1. Set up a formula for V using the symbols in **figure 8** for the indicated dimensions of the tube.

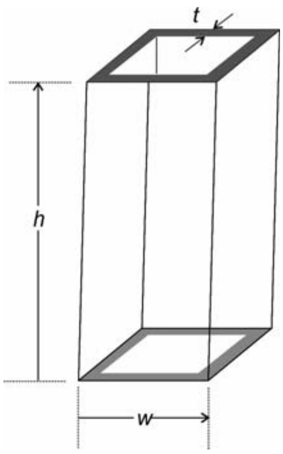


Fig. 8 The dimensions of the metal tube.

2. Measure each of the dimensions of the tube several times and calculate the average. Try to obtain as large a range of values for each of the dimensions as you possibly can. See to it that you read the vernier scale off correctly for each of the measurements. Convert all your measurements to S.I. units and tabulate them.

3. Measure the mass of the beaker on the electronic scale. Check that the scale is level and zoered. Calculate the density of the metal. Round the value off to the correct number of significant figures.

4. Look in **Table 1** for materials with similar densities. What can you conclude from this?

PAR

Projectile motion

Objectives

In this experiment some of the features of projectile motion will be investigated, like

- The path or trajectory followed by a horizontally-launched projectile.
- The relationship between the coordinates of its trajectory.
- A simple check will be made to see whether the results obtained are consistent with the theory of projectile motion.

Theory

While considering the motion of the moon around the earth Newton realised that it was falling around the earth just like a cannonball would if such a ball was shot fast enough from the top of a high mountain - see **figure 1**. The force responsible for such motion he formulated in his universal law of gravitation. This law states that there exists a force, F_G, between any two objects in the universe. The force is directly proportional to the product of the masses, m_1 and m_2, of the objects and it is inversely proportional to the square of the distance, d, between their centres of mass, that is

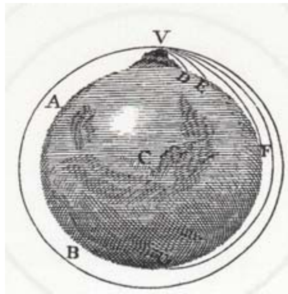


Fig. 1 Putting a cannonball into orbit.

$$F_G = G\frac{m_1 m_2}{d^2}$$

The proportional constant $G = 6{,}67 \times 10^{-11}\ m^3/kg \cdot s^2$ is known as the universal gravitational constant. This force of gravity tends to pull the objects towards each other. In **figure 2** we see how this gravitational force pulls the moon out of its straight line path into an orbit around the earth.

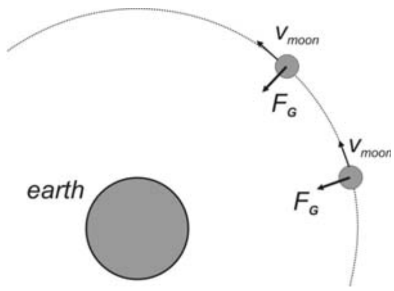


Fig. 2 The orbit of the moon around the earth.

The earth generates a gravitational field around it due to this force as depicted in **figure 3**. The strength of the gravitational field at a particular point in space, g_G, is the force that a unit mass will experience at that point in the field and is given by

$$\begin{aligned} g_G &= \frac{F_G}{m} \\ &= G\frac{Mm}{md^2} \\ &= G\frac{M}{d^2} \end{aligned}$$

where M is the mass of the earth, $5{,}98 \times 10^{24}\ kg$, and m is the mass of the object in the gravitational field. The strength of the field at the surface of the earth ($d = 6370 \times 10^3\ m$), for which we use the symbol g, is therefore

$$\begin{aligned} g &= 6{,}67 \times 10^{-11}\,\frac{5{,}98 \times 10^{24}}{(6370 \times 10^3)^2} \\ &= 9{,}83\ N/kg \end{aligned}$$

Note that it follows from Newton's second law that the strength of the field is also the acceleration that an object in free-fall will experience at that point in the field

$$\begin{aligned} F_{net} &= m\,a \\ m\,g_G &= m\,a \\ \therefore\ g_G &= a \end{aligned}$$

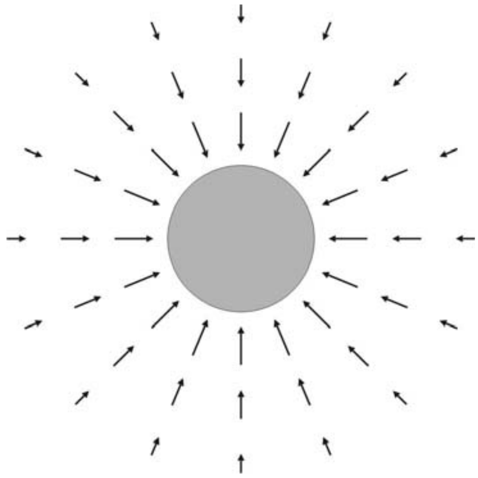


Fig. 3 The gravitational field of the earth.

The gravitational field weakens according to an inverse square relationship as the distance from the earth increases, but we consider it to be constant over the surface of the earth since the field decreases by only one percent at an altitude of about thirty kilometers above the surface of the earth. We will also consider the field to be parallel (directed downwards) over the course of the trajectory of the projectile althought it is slightly convergent towards the earth's centre of mass.

Consider a projectile launched horizontally with an intial velocity $\vec{v}_o$ at a height h_o above the ground - see **figure 4**. The position of the projectile is drawn at equal time intervals. The projectile would have followed path A if the gravitational field was not present , but due to its presense we find that the downwards speed of the projectile, v_y , increases at a constant rate of about $9{,}8\,m/s$ each second. The resultant velocity of the projectile, $\vec{v}$, shows in which direction the projectile moves at each point. Note that the horizontal speed of the projectile is unaltered by the gravitational field. The motion of the projectile is therefore a combination of a horizontal motion with a constant speed and a vertical motion with a constant acceleration downwards.

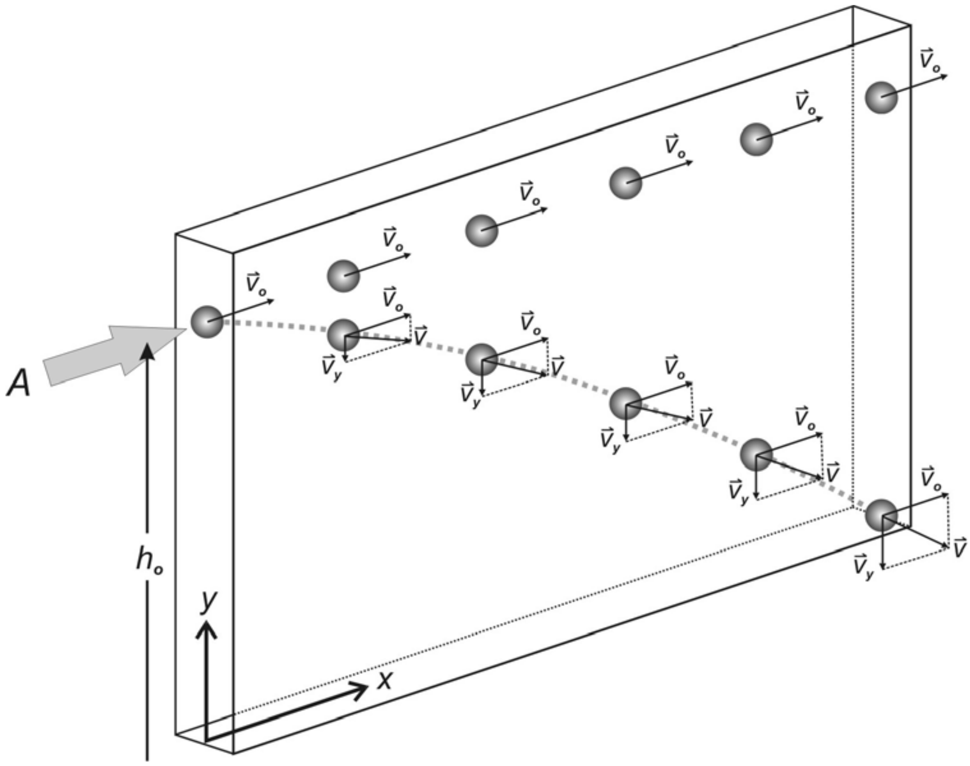


Fig. 4 The motion of a horizontally launched projectile in a gravitational field.

Consider a coordinate system at floor level with the origin directly below the point where the projectile is launched, and orientated as shown in **figure 4**. The x-coordinate of the projectile is given by

$$x = v_o t \quad \text{....................} \ \textbf{(1)}$$

where t is the elapsed time since it was launched.

The y-coordinate of the projectile when it is launched is $y_o = h_o$. The projectile drops further and faster below this level as it moves forward due to its increased downward speed. The distance, *d*, that it drops downward from this level can be calculated with

$$d = \tfrac{1}{2} g t^2$$

since the initial downward speed is zero. The y-coordinate of the projectile is then

$$\begin{aligned} y &= y_o - d \\ &= y_o - \tfrac{1}{2} g t^2 \quad \textbf{.............................. (2)} \end{aligned}$$

From **equation (1)** we have that

$$t = \frac{x}{v_o}$$

If we substitute this into **equation (2)** we have that

$$y = y_o - \frac{g}{2v_o^2} x^2 \quad \textbf{........................... (3)}$$

which gives us the expected relationship between the coordinates of the projectile. We will use the data of this experiment to verify this relationship. Notice that **equation (3)** describes a parabola with the y-axis as its axis of symmetry and its “legs” pointing downwards. Taking into account that the gravitational field is slightly convergent and not parrallel will result in an elliptical relationship between the coordinates. That is if the mass of the earth could have been concentrated at its centre of mass we would find that a projectile would actually have followed an elliptical orbit about this concentration of mass.

For your interest - a bit of advanced physics

A small ball with a diameter of 25 cm is used in this experiment for the projectile. Its speed will be somewhat less than that we expect it to be due to drag that it encounters in flight. This causes it to deviate from the expected trajectory especially towards the end of the flight due to an increase in its speed. We also find that a projectile can be deflected from its path if it is spinning. This deflection can occur in any direction - up, down or sideways depending on the relationship between the orientation of the axis of rotation and the direction of its motion. Consider a ball that is moving to the right with its axis of rotation pointing out of the paper - see **figure 5** . Suppose it is spinning anticlockwise about the axis. The air is flowing over the projectile from right to left as it moves forward. The ball drags a thin layer of air with as it spins around. This thin layer then impedes the airflow over the ball somewhat at the bottom and speeds it up at the top. According to Bernoulli's theorem the air pressure at the bottom will then be slightly higher than the pressure at the top. The ball is therefore pushed upwards into the faster moving air stream due to this pressure difference. This is called the Magnus effect.

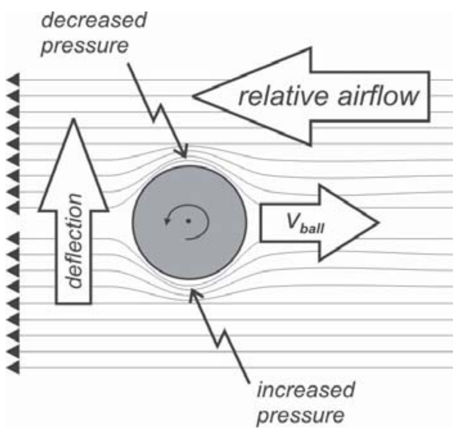


Fig. 5 A moving spinning ball pushed aside by the Magnus effect.

The projectile launcher used in this experiment is designed not to cause the ball to spin when it exits the barrel. The ball rests inside a recess in a piston which slides in the barrel of the launcher - see **figure 6**. Please note that you should check through the side windows that the ball is resting in the piston rather than lying in the barrel before each launch. **Never look into the front of the launcher - always look through the little windows along the side of the barrel!**

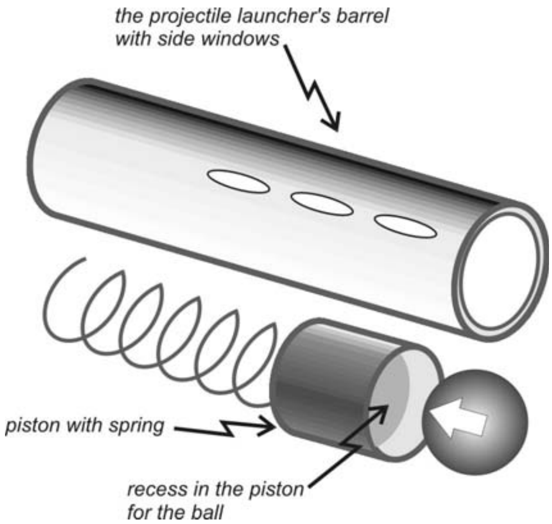


Fig.6 A simplified view of the piston after being removed form the barrel of the launcher.

Setup

1. Clamp the base plate of the projectile launcher to the seat of a laboratory chair with a C-clamp. Align the front of the launcher with the edge of the seat - see **figure 8**. Put the chair in the middle of the aisle between the laboratory benches with the launcher facing the window. Put a heavy mass piece on the seat of the chair.

2. Put a piece of paper on the floor just in front of the chair and stick it to the floor with some masking tape.

3. Adjust the barrel of the launcher so that it is horizontal, that is so that the little plumb is hanging along the 0^{o} line (no elavation) - **figure 7a**. Tighten both bolts on the other side of the launcher - **figure 7b**.

4. Use the plumb line to mark on the paper the point where the ball leaves the barrel. This will be the origin of the coordinate system - see **figures 8 and 9**.

5. Place the ball into the front of the launcher and push it back with the ramrod to the long range setting.

6. Check that nobody is in the path of the projectile and launch it. Determine where the ball lands and

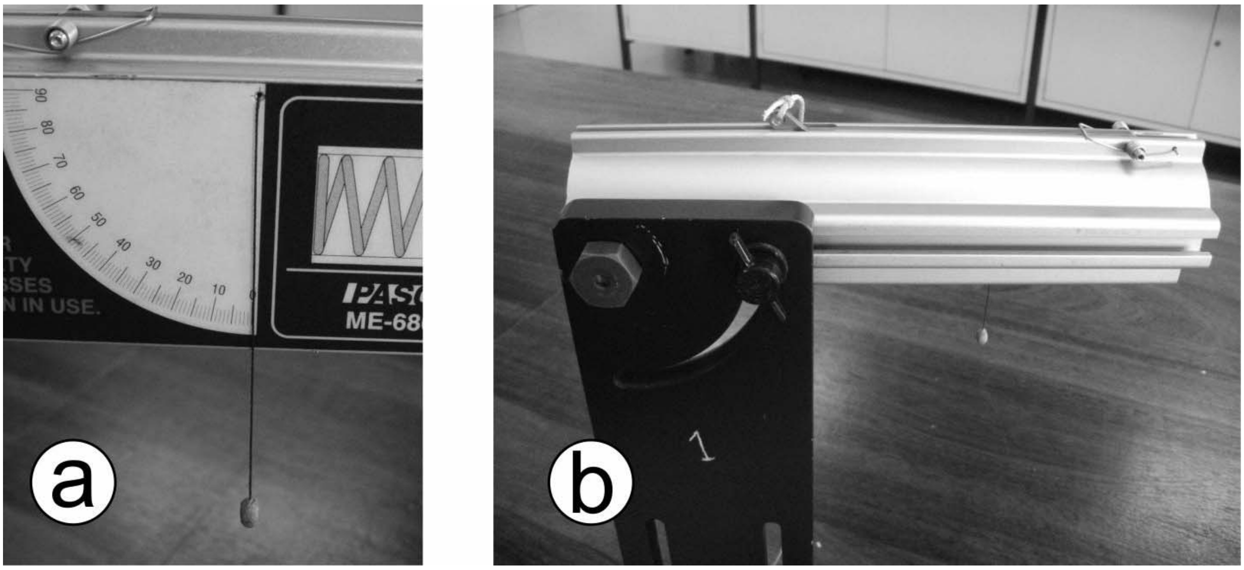


Fig. 7 Preparation of the launcher before shooting - see setup note number **3**.

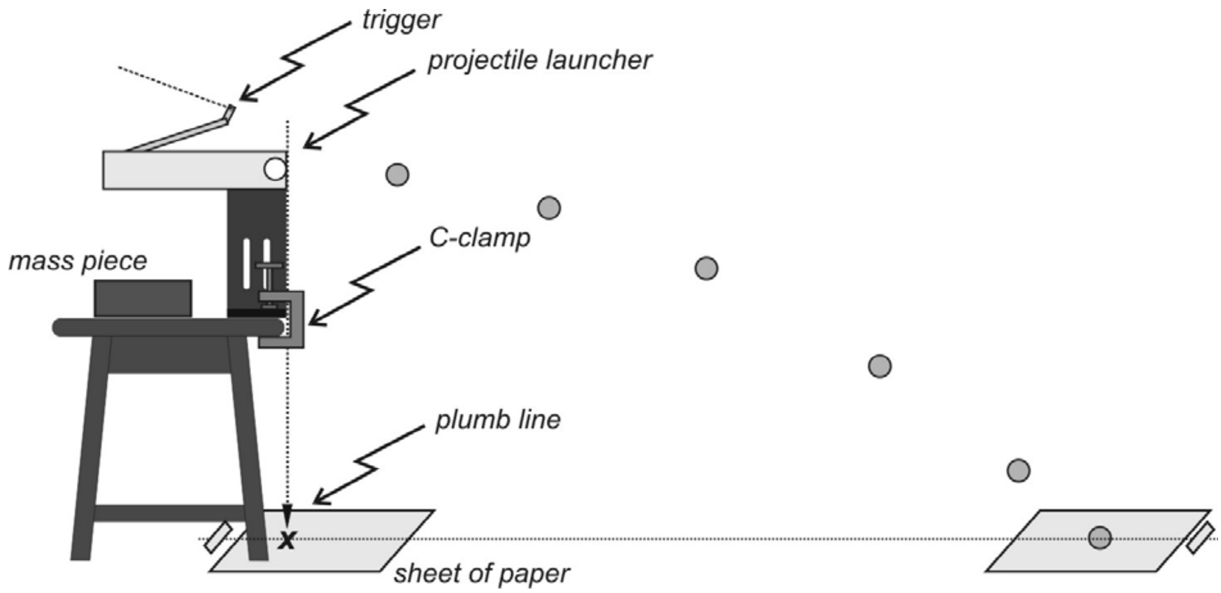


Fig. 8 The setup for the projectile experiment.

stick another sheet of paper to the floor at this point. Put an A4 sheet of carbon paper on top of it.

Determining v_o

1. Launch the ball several times and determine the average distance to the point where it lands on the floor. Check that the settings in **points 3** and **4** in the **Setup** notes still holds before each launch.

2. Add half the diameter of the ball to this value and note it down as x_t (touchdown) in your report book (we are mapping the position of the front of the ball in this experiment).

3. Determine the height through which the ball drops and calculate v_o. Show your calculations.

The coordinates of the ball along the trajectory

Read through all the points below before starting with the measurements.

- Use a wooden stand to determine the position of the front of the ball at five points along the trajectory. Place the stand in the path of the ball and use the small piece of carbon paper on top of another piece of white paper to find the point where the ball strikes it. Increase the horizontal distance with the same amount for each measurement.
- Tabulate the coordinates of the front of the ball, *P* **figure 9**.
- Include the coordinates of the ball where it leaves the barrel and lands on the floor in your table.
- Remember to **first insert** the ball in the launcher **before** cocking it.
- Never look into the front of the barrel!
- Check that the settings in **points 3** and **4** in the **Setup** notes still holds before each launch.

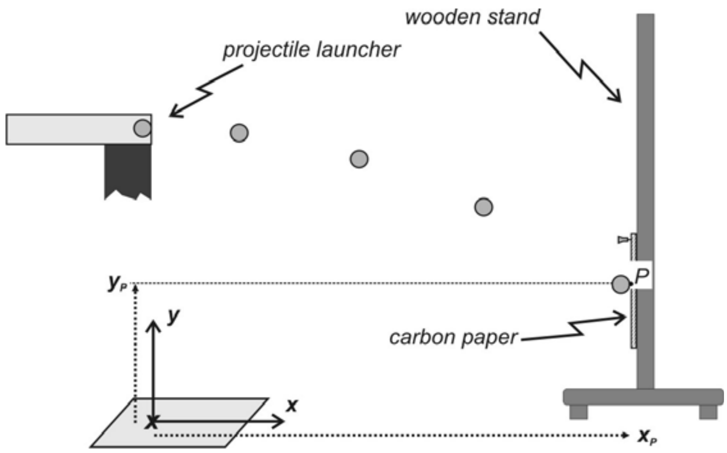


Fig. 9 The coordinates of the point *P* of the front of the ball along the trajectory.

Analysis

1. How would the coordinates of the middle of the ball, in a coordinate system with the origin moved backwards to the point below the middle of the ball where it leaves the piston, differ from the coordinates you obtained in this experiment? Explain.

2. Draw a graph of the *x*- and *y*-coordinates of the ball. Use the same scale on both axes.

3. Draw another graph that indicates the relationship between the y-coordinates and the square of the *x*-coordinates.

4. What is the possible relationship between the *x* and *y*-coordinates of the ball?

5. Write the empirical relationship between the coordinates down. Supply values and units for the constants in the equation.

6. Use the gradient of the graph to determine another value for v_o.

7. Calculate the percentage difference between the two values.

Safety precautions

Never look into the front of the launcher, you risk getting shot in the eye! Look through the little windows along the side of the barrel if you need to see the cocking position or observe the ball.

ELA

Elasticity

Study aims

You must be able to:

1. Investigate the elasticity of a steel spring and a rubber band.

2. Draw a graph to demonstrate the relationship between the applied load force and the extension of a spring .

3. Use Hooke's law to find the force constant, *k*, of a spring.

Introduction

The elasticity of a body is the ability of the body to return to its original size and shape after being distorted by some force. If, after distortion, the body recovers its size and shape completely, it is said to be perfectly elastic. From everyday observations we know that some objects, like helical springs and rubber bands stretch when a force is applied to them. When the force is removed, the objects appear to return to their original shape.

Hooke's law

When a spring is stretched in the y-direction, it exerts a **restoring force** $\vec{F}$ (elastic force) which is directly proportional to the extension $\vec{y}$. The relationship between the restoring force and the extension is given by **Hooke's law**:

$$\vec{F} = -k\vec{y} \text{ .... } \mathbf{(1)}$$

k is the **force constant**. The minus sign indicates that the restoring force **opposes** the extension. If we are only interested in the magnitudes of the quantities then we can simplify this relationship to:

$$F = k\,y \text{ ....}\mathbf{(2)}$$

At equilibrium we find that the load force *W* is the same as the restoring force *F*, therefore:

$$W = k\,y \text{ ....}\mathbf{(3)}$$

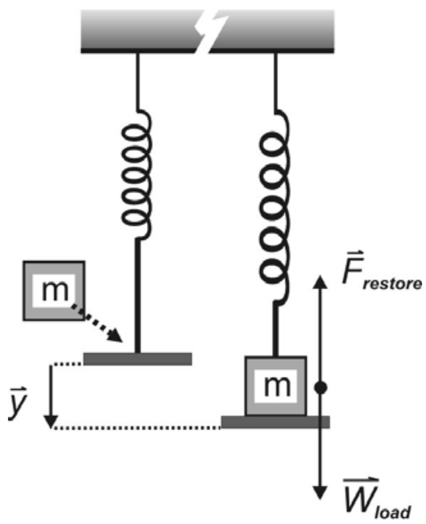


Fig. 1 The extension of a spring by a load force.

Experiment

A. Elasticity of a helix spring

1.Suspend the mass hanger, with a mass m_0, from the spring.

2. Take the reading y_0 on the ruler by using the bottom of the hanger as a reference.

3. Load the spring in steps of 50 g masses and note the corresponding y' readings on the ruler. Do not exceed a total load of 250 g (excluding the mass, m_0, of the hanger).

4. Remove the masses from the hanger one by one and again take y' readings of the extension of the spring on the return path. Does the spring return to its original length in each case?

5. Set up a table of **extension** $y = (y' - y_0)$ against the **load** m, where m is the **total mass** placed on the hanger. Note that m does not include the mass m_0 of the hanger!

6. Calculate the load forces, W, exerted by the different masses using the relationship $W = mg$. Take $g = 9{,}78\ \text{ms}^{-2}$.

7. Draw a graph of the extension, y - the dependent variable, of the spring as a function of the load force, W - the independent variable, applied to it.

8. Does the spring obey Hooke`s law?

9. Set up the empirical relationship between y and W. Also supply a value, and units, for the proportionality constant in the equation.

10. Change the subject of **equation 3** to y. Compare this equation to your empirical equation. What is the relationship between the proportionality constant and the force constant, k, of the spring?

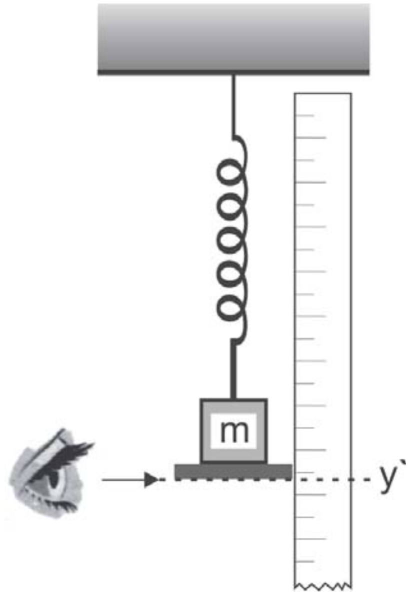


Fig.2 Measuring the position of the extension of the spring.

11. Determine k, and its units, from the proportionality constant. Round the value off to a realistic number of significant figures.

12. Suspend the "unknown" mass from the spring and measure the extension of the spring. Use **equation 3** to calculate the mass of the "unknown" mass.

B. The elasticity of a rubber band

1. Replace the spring with a rubber band.

2. Repeat steps 1 to 8 for the rubber band. Support the mass hanger in your one hand while you place the masses on the hanger with your other hand. Slowly lower your hand after placing each mass unit on the hanger. Wait until the rubber band has stabilised before taking a reading. When removing a masspiece it is advisable to hold the top one and remove it slowly from the pile.

3. Draw the extension - load force graph of the data on a new graph. Indicate on the graph in which direction the extension and contraction progressed.

Questions

1. Are there any differences in the elasticity of the rubber band and the helical spring? If so, describe the differences in detail (there are at least three).

2. Do you think Hooke's law can be used to describe the elasticity for both rubber and steel (assuming that the different materials have different k values)? Explain.

3. If you were to design a spring scale, would you prefer to use a steel spring or a rubber band? Why?

Further Investigations

C.L. Stong. "Some Delightful Engines Driven By the Heating of Rubber Bands" *Scientific American*, April 1971

Safety precautions

Be careful with the spring and elastic. It could shoot you in the eye if it suddenly comes loose, especially when it is under tension.

SHM

Simple harmonic motion

Introduction

We find many examples of oscillating systems in everyday life. Watches use the precise oscillations of quartz crystals to keep time accurately. Information is encoded into the oscillations of electromagnetic waves and then transmitted to satellites which beam the signal back down to earth. Sound waves are propagated via oscillating air molecules.

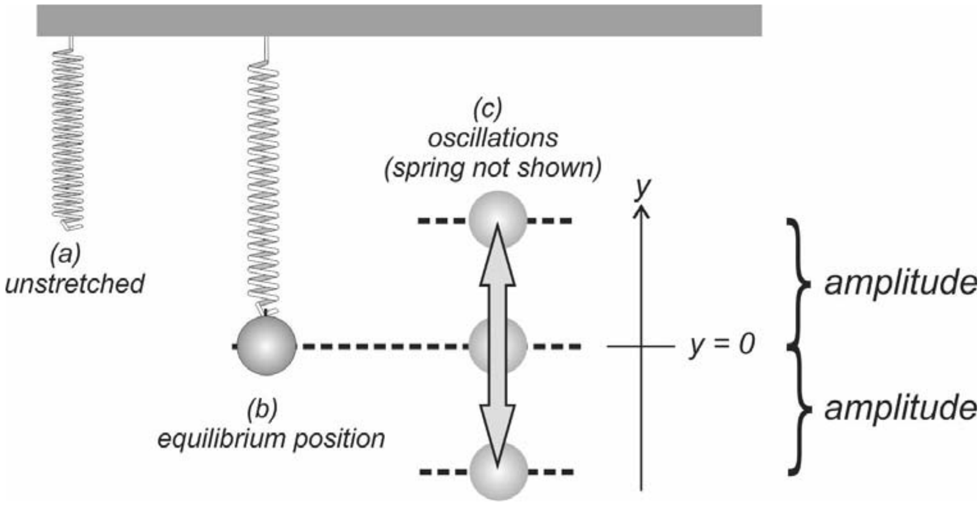


Fig .1 An oscillating mass on a spring.

In this experiment we will investigate the oscillations of a mass piece hanging on a spring, **figure 1.** The *y*-position of the mass piece is a function of time, $y_e(t)$ (*e* - experimentally determined), and we will calculate these positions from the restoring force of the spring, F_R, as measured by a force sensor connected to a computer - **figures 2** and **3**. We will then compare these positions to those calculated for simple harmonic motion, $y(t)$, a special category of oscillations. These two sets of y-positions should be very much the same if the oscillating mass piece exhibits simple harmonic motion.

Fig. 2 Spring hanging from the force sensor.

Theory

We will assume that the restoring force of the spring obeys Hooke's Law - see **figure 3**

$$F_R = ky \quad \text{..................... (1)}$$

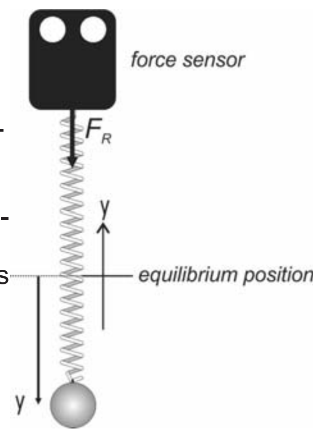
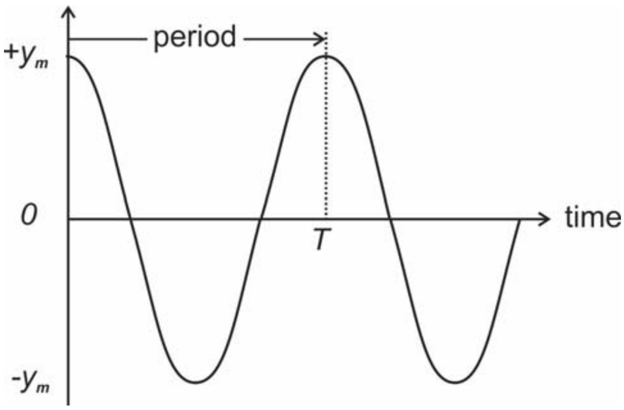


Fig. 3 The restoring force of the spring.

The force sensor will be zeroed with the mass piece hanging in the equilibrium position. That, together with the fact that it displays downward-acting forces as negative leads to the omission of the negative sign normally found in **equation 1** (the restoring force at the bottom end of the spring acts in the opposite direction to that which the force sensor measures at the top). The *y*-positions of the mass can then be calculated if the force constant, *k*, of the spring is known

$$y_e = F_R / k \quad \text{..................... (2)}$$

The y-positions of simple harmonic motion are depicted in the graph below as a function of **time**. y_m is the amplitude of the oscillations. We will use the extreme positions of the mass about the equilibrium position, $y = 0$, for the value



of the amplitude - see **figure 1**. As you can see this graph is a basic cosine function

$$y = y_m \cos(\theta) \text{ .... } \mathbf{(3)}$$

θ is called the phase of the motion. It is time-dependent and is given by

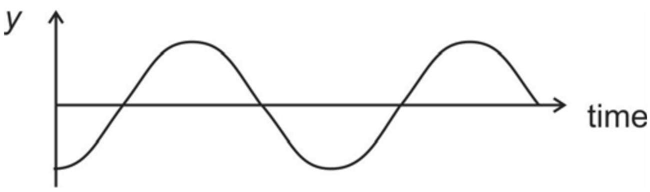
$$\theta = \omega t \text{ .... } \mathbf{(4)}$$

where t is time and ω is the angular frequency. The angular frequency is a constant that determines the rate at which the oscillations occur, and is measured in radians per second (see the section at the end of the notes for some information on the radian angular measure). The angular frequency, ω, is dependent on the period of the oscillations, T (the time to complete one cycle) - see the above graph. It is given by

$$\omega = \frac{2\pi}{T} \text{ rad/s .... } \mathbf{(5)}$$

It follows from **equation 4** that $\theta = 2\pi$ radians when $t = T$, that is, the phase of an oscillation or cycle is 2π radians and the y position at this point is the same as that at $t = 0$ - see the graph above.

In general we would not have that $y = +y_m$ at $t = 0$. Consider the following three cases of simple harmonic motion:

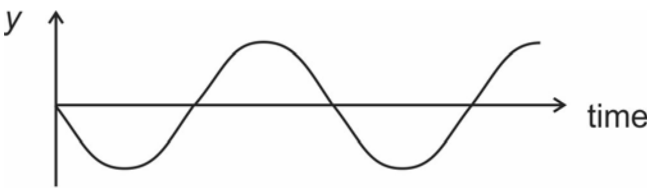


1. At $t = 0$ the y-position $-y_m$ in this case. **Equation 3** needs to be modified to describe this new graph and the way we do it is to add an angle ϕ called the phase angle or phase constant so that the phase is in general given by

$$\theta = \omega t + \phi \text{ .... } \mathbf{(6)}$$

In this case $\phi = \pi$ radians so that the equation of the graph is

$$y = y_m \cos(\omega t + \pi)$$

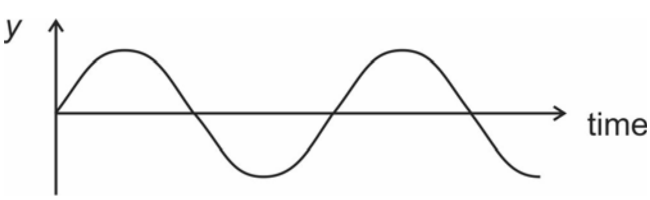


2. Now we have that $y = 0$ at $t = 0$. It follows from **equations 6** and **3** that

$$\cos(\phi) = 0$$
$$\therefore \phi = \pi/2 \ rad$$

Check it with your calculator set to radians. We now have for this graph that

$$y = y_m \cos(\omega t + \pi/2)$$



3. Here we again have that $y = 0$ at $t = 0$, but now the graph continues along a different path - upwards instead of downwards as in the previous case. Note that due to the symmetry of the cosine function - see **figure 4** - that both the positive and the negative angles are possible solutions to the equation

$$\cos(\phi) = 0$$
$$\therefore \phi = \pm\pi/2 \; rad$$

Try it with your calculator. So which solution is the correct one or will both do?

According to **equation 6** we have that at $t = 0$ that $\theta = \phi$ and as time progresses the value of θ will steadily increase from this angle onwards. Looking at **figure 4** (note the x-axis is θ) we see that if we were to choose the solution $\theta_o = \phi = \pi/2 \; rad$, point a, then the graph or oscillations, would continue downwards, but if we take the solution $\theta_o = \phi = -\pi/2 \; rad$, point b, then the graph will continue upwards. So in case 3

$$y = y_m \cos(\omega t - \pi/2)$$

is the correct formula for the graph.

The y-displacement formula for simple harmonic motion is

$$y(t) = y_m \cos(\omega t + \phi) \; \textbf{..........(7)}$$

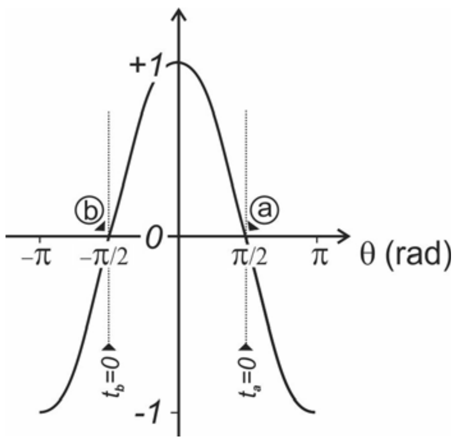


Fig. 4 The cosine function.

We will have to determine the values of y_m, ω and ϕ that go with the oscillating mass in order to calculate *y(t)* from it.

When a spring drives the motion of the mass, with mass *m*, the simple harmonic theory indicates that the angular frequency, ω, can also be calculated from the force constant, *k*, of the spring according to

$$\omega^2 = \frac{k}{m} \; \textbf{...............................(8)}$$

More information can be found in your textbook in chapter 15.

Apparatus

A force sensor is used in this experiment to measure the restoring force that drives the motion of the masspiece - see **figure 2**. It uses a strain gauge to measure the force. The sensor connects to a computer via an interface box - see **figures 5 and 6.** It can measure a push or a pull acting on it up to 50 N. The sensor registers a downward pull as a negative force and an upward push on it as positive force. It can be zeroed by pushing the Tare button on the <u>sensor</u>. The Science Workshop program on the computer reads the sensor via the interface box when it is in either the **MON**itoring or **REC**ording mode - see **figure 7**.

Fig. 5 The front of the computer interface.

Fig. 6 The rear of the computer interface.

Apparatus setup

1. Mount the force sensor on the rod on the laboratory stand with its hook pointing downwards - see **figure 2**.

2. Plug the sensor into channel A of the interface box - **figure 5**. The indentation on the metal rim of the plug should be towards the *TOP*.

3. Check that the interface box is switched on and plugged in - **figures 5** and **6**.

4. Open the program file on the computer as instructed by the demonstrator. The program reads the sensor when it is put into either the monitoring (**MON**) or the re-

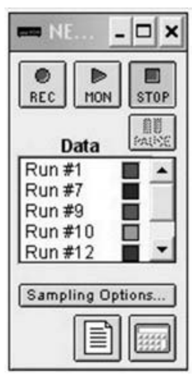


Fig. 7 The program SetUp window.

cording **(REC**) mode - **figure 7**.

5. Hang the spring on the hook of the sensor and suspend the mass hanger from it with no mass pieces on it. Loosen and then tighten the thumb screw on top of the force sensor. Zero the sensor by pressing the Tare button on the sensor while the hanger is stationary.

Determining the force constant of the spring

Fig. 8 The Digits display window.

Use the monitoring mode - **MON figure 7** - of the program to measure the restoring force while you stretch the spring downwards by a known amount. The force values appear in the Digits display window. Calculate k in N/mm. Use the average of several measurements for your calculation. **STOP** the monitoring mode when you are done and remove the hanger from the spring.

<u>Notes:</u>

- Do not stretch the spring more than about 10 cm.
- Zero the sensor with the mass hanger hanging stationary.

Recording several oscillations

Replace the mass hanger with the mass pieces that are fixed together - the oscillator. Record a few oscillations of the oscillator with the program. Call the demonstrator to check your recording before proceeding with the rest of the experiment.

<u>Notes:</u>

- Zero the sensor with the oscillator hanging stationary in the equilibrium position ($y = 0$).
- First set the oscillator in motion before starting the recording.
- Keep the amplitude of the oscillations small - a few centimeters at most.
- The oscillator should not swing sideways while you are recording its motion.
- Recordings can be deleted by clicking on their runs - **figure 7** - and pressing the Delete key on the keyboard.

The *y*-displacements of the oscillator

Next you are going to calculate the experimentally-determined y-displacement, y_e, of the mass at several points in the first recorded oscillation.

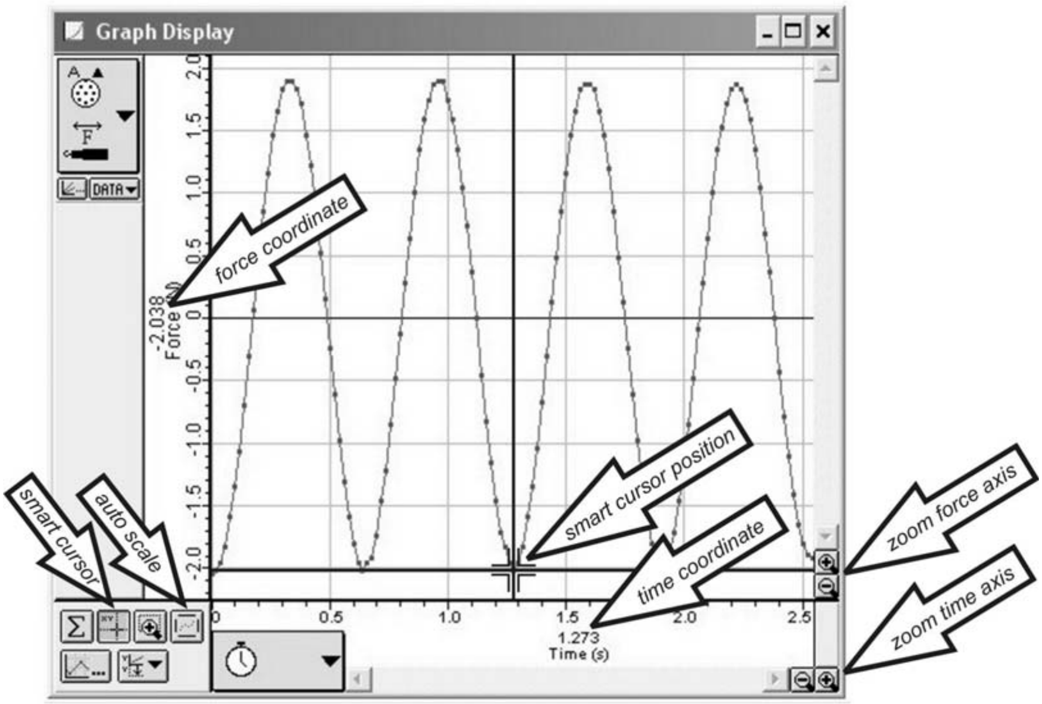


Fig. 9 Recorded oscillations as displayed in the Graph Window of the program.

Maximise the Graph Window and click the autoscale button in it - see **figure 9**. The graph displays how the restoring force of the spring, F_R (the dependent variable), changed with time, t (the independent variable).

Drag the cursor over a datapoint in the graph. The time and force coordinates of the chosen datapoint are selected in the table window - **figure 10**.

Select six datapoints in the first oscillation of the graph. Try and space the chosen datapoints evenly throughout the oscillation. Calculate y_e in millimeters for each of the datapoints and tabulate your results.

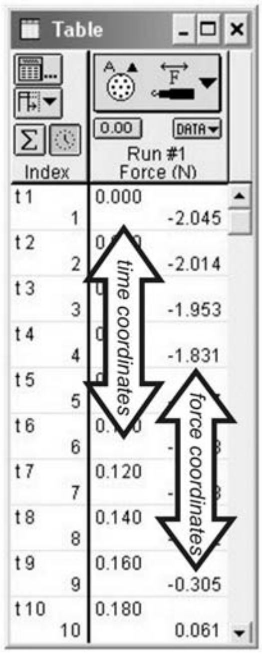


Fig. 10 The Table window of the program.

Determining the angular frequency and amplitude

Click on the smart cursor in the Graph Window - **figure 9** - and move it around in the graph area. Note how the force and time coordinates of the cursor are displayed next to the Time(s) and Force (N) axis labels of the graphs - also indicated in **figure 9**.

Use the smart cursor to take readings on the graph to detemine the period, T, of the oscillations. Take several readings and calculate an average period. Make a rough sketch of the graph and indicate on it where you made your readings.

Calculate the angular frequency of the oscillations from **equations 5** and **8**. The values should be almost the same. Use the average of the two values for the *y*- displacement calculations you are going to make later on.

Use the smart cursor to read off the minimum and maximum restoring force values, $F_{\min}$ and $F_{\max}$, and calculate the amplitude, y_m, of the oscillations in millimeters from these.

Determining the phase constant

Again use the smart cursor to read off F_o, the restoring force of the spring when $t = 0$. Calculate y_o, the displacement of the oscillator when $t = 0$ from it and subsequently determine the phase constant, ϕ (in radians).

The y-displacements of the simple harmonic oscillator

Set up the simple harmonic motion *y*-displacement equation with values and units, in brackets, for the constants in the equation. Use this formula to calculate the *y*-displacements of the simple harmonic oscillator, *y(t)*, at the same stages as the six y_e values you chose previously. Calculate and tabulate the *y(t)* values, in millimeters, together with the difference between the two displacements $\Delta y = y_e - y$.

Conclusion

Did the mass exhibit simple harmonic motion during the first recorded oscillation? Explain.

Radian angular measure

Just as we can measure distances in different measures like kilometres or miles, for example, the distance between Johannesburg and Pretoria is about 53 km or 33 miles, so angles can be measured in degrees or radians. Here we have that 360° is 2π radians, so that 40° is converted to radians as follows:

$$360^o = 2\pi \ rad$$

$$\therefore 40^o = 40^o \times \frac{2\pi \ rad}{360^o} = 2\pi/9 \ rad = 0{,}698 \ rad$$

and

$$1 \ rad = 1 \ rad \times \frac{180^o}{\pi \ rad} = 57{,}3^o$$

See http://www.mathsisfun.com/geometry/radians.html for more information.

Safety precautions

- Be careful of the supporting rod on the laboratory stand. It could poke you in the eye. Especially when it is pointing towards you.
- Be careful with the spring. It could shoot you in the eye if it accidently comes undone.

CLM

Linear momentum

Introduction

In this experiment you will arrange a collision between a moving flat disc *S* - the striker, approaching from the lower right hand corner in the diagram and a smaller flat disc *T* - the target, lying stationary in the middle of a horizontal flat board - **figure 1**. You will then investigate whether the linear momentum of the system of discs was conserved in spite of the collision.

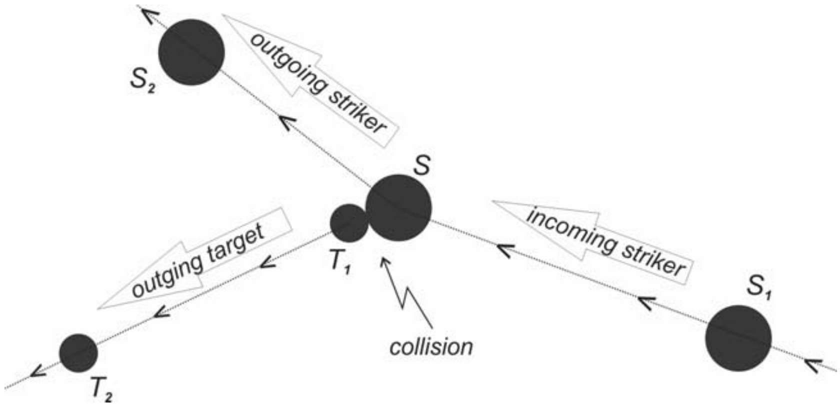


Fig. 1 Disc *S* collides with a smaller stationary disc *T*.

Theory

In **figure 2** we have a smooth stone, *a*, sliding along on a level flat frictionless surface, like ice. Its linear momentum, $\vec{p}_{a1}$, is represented by the arrow. The arrow is pointing in the direction in which the stone is sliding. The direction of the momentum vector is θ_{a1}, measured anticlockwise from the positive x-axis.

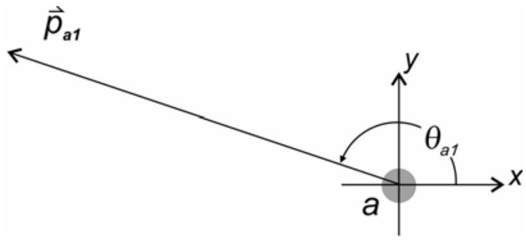


Fig. 2 The momentum vector of a stone, *a*, sliding on a level flat frictionless surface.

According to Newton's first law the velocity of the stone will remain unchanged as it slides along on the surface. The momentum of the stone will therefore stay the same in magnitude and direction, unless a non-zero net force acts on it. The magnitude of the momentum of the stone is its mass times its speed

$$p_{a1} = m_a v_{a1}$$

It determines how long the arrow in the diagram should be drawn. For example if the momentum of the stone is $1{,}2\,kg\,m/s$ and the scale is $30\,mm\ /\ 1kg \cdot m/s$ then the arrow should be drawn with a length of

$$1{,}2\,kg\,m/s \times \frac{30\,mm}{1\,kg\,m/s}$$
$$= 36\,mm$$

thirty six millimeters. One should choose a scale such that the length of the arrow has a reasonably long length when drawn on the paper. This will impove the accuracy of subsequent vector manipulations.

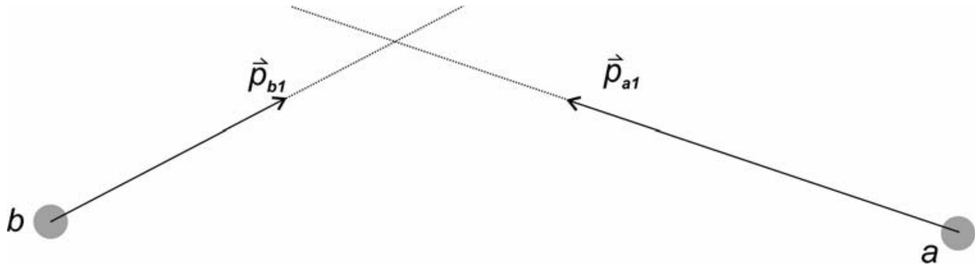


Fig. 3a The momentum vectors of two stones on a collision course drawn to scale.

In **figure 3a** we have another stone sliding along on the surface and it is on a collision path with the first stone. We can combine the momentum vectors of these two stones as depicted in **figure 3b** giving the resulting momentum vector $\vec{p}_1$. $\vec{p}_1$ is drawn from the tail of the first vector to the head of the second vector after positioning the two original arrows head-to-tail. $\vec{p}_1$ is then the representation of the combined or resultant momentum of the two stones before the collision. Symbolically we write it as

$$\vec{p}_1 = \vec{p}_{a1} + \vec{p}_{b1}$$

where the + symbol refers to vector and not normal addition. Note that the order in which the two vectors $\vec{p}_{a1}$ and $\vec{p}_{b1}$ are added will not change the resultant vector, that is

$$\vec{p}_{a1} + \vec{p}_{b1} = \vec{p}_{b1} + \vec{p}_{a1}$$

This resultant momentum of the two stones, $\vec{p}_1$, will remain the same up to the point where they collide. The question is how will the collision change the resultant momentum of the two stones? That is what will their resultant momentum be after the collision?

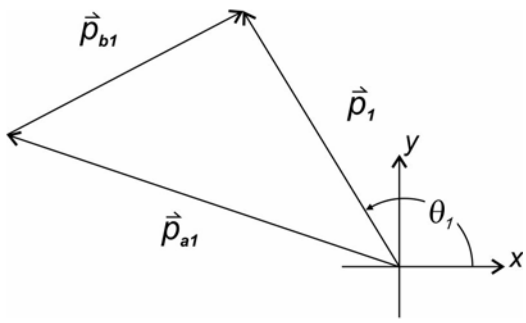


Fig. 3b The resultant momentum, $\vec{p}_1$,of the two stones before the collision drawn to scale.

Figure 4a show the two stones at the point of impact. The collision force $\vec{F}_{ba}$ is the force stone *b* exerts on stone *a*. This force acts on stone *a*. $\vec{F}_{ab}$ is the force stone *a* exerts on stone *b*. It acts on stone *b*. These collision forces change the momenta of both stones after the collision, which is shown in **figure 4b**. How is the resultant momentum of the two stones changed by the collision? The answer lies in looking at the collision forces. We saw that a non-zero net force is necessary to change momentum. This is also true for the combined or resultant momentum of the two stones, but according to Newton's third law the two collision forces are of the same magnitude but act in opposite directions on the stones, that is

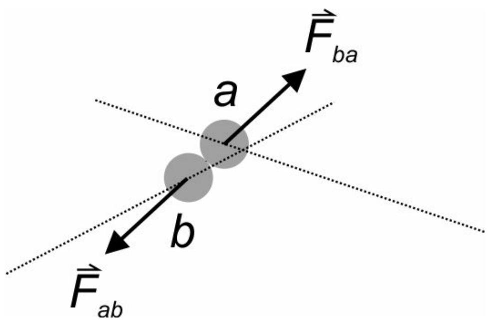


Fig. 4a The collision forces acting on the stones at the point of impact.

$$\vec{F}_{ba} = -\vec{F}_{ab}$$

The resultant of these two forces is therefore zero. This means that the resultant momentum of the two stones will not be changed by the collision or

$$\vec{p}_1 = \vec{p}_2$$

or in words

resultant momentum before the collision = resultant momentum after the collision

We say that the resultant momentum of the system of two stones is conserved, or stays the same. In conclusion we find that although the collision forces change the individual momenta of the stones their resultant or combined momentum is left unchanged by the collision forces since they cancel each other out. This is only true if no other external net force acts on the system.

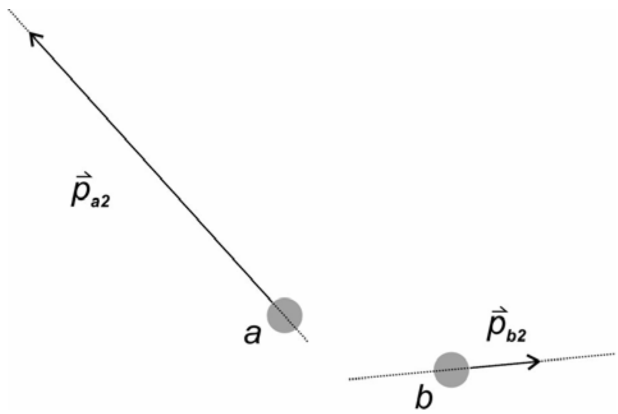


Fig. 4b The momenta of the stones after the collision drawn to scale.
Compare this with **figure 3a**.

Apparatus

- One large wooden board with a piece of paper taped on top of it. The board should have no dirt on it and the paper should be as flat and tight as possible.
- Two round plastic discs, one large and one small. The large disc is placed on the paper with the grooved side downwards. One side of the small disc is machined flat. This side should be placed downwards on the paper.
- A wooden ruler for calibration purposes.
- A protractor used to draw the vector arrows in their measured directions in your book.
- An elastic band that is used to shoot the large disc from one of the short sides of the board towards the smaller stationary disc that is placed in the middle of the board - see **figure 5**.
- A digital camera to record the collision and a cable to download the videos onto the computer.
- A lamp mounted above the board.
- A computer with the Tracker program installed on it.

Filming the collision

1. Fix a sheet of paper to the top of the large wooden board.

2. Place the objects on the paper as in **figure 5**.

3. Mount the camera above the wooden board so that it is looking downwards towards it.

4. Slide the board so that the small disc is positioned below the lens of the camera. Make sure that the ruler is included in the camera's image.

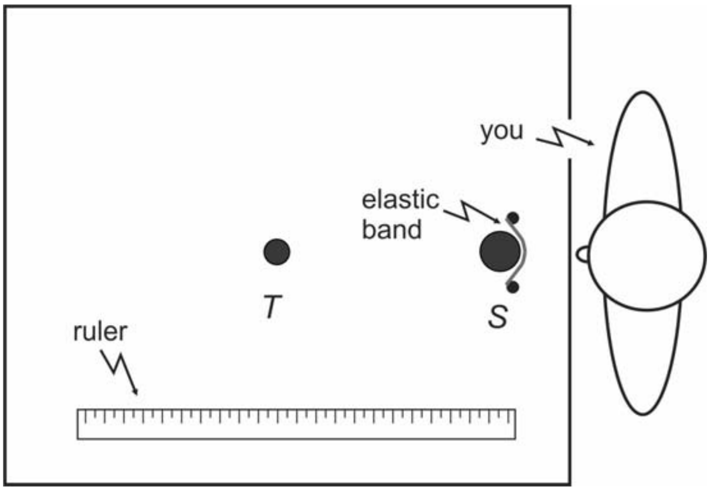


Fig. 5 Arranging the objects on the sheet of paper.

5. Switch the lamp on.

6. Pratice shooting the small disc with the larger one so that both discs move off at an angle after the collision.

Question 1 Do the discs travel with a constant speed along their tracks?

Question 2 Do you therefore expect that momentum will be conserved completely? Why?

Question 3 Can you identify another external force that will also alter the resultant momentum of the discs? Name it.

Question 4 Explain why we should measure the speed of the discs just before and just after the collision and not where they are far away from the point of impact.

7. Make videos of a few succesful collisions.

Getting ready to analyze the collision

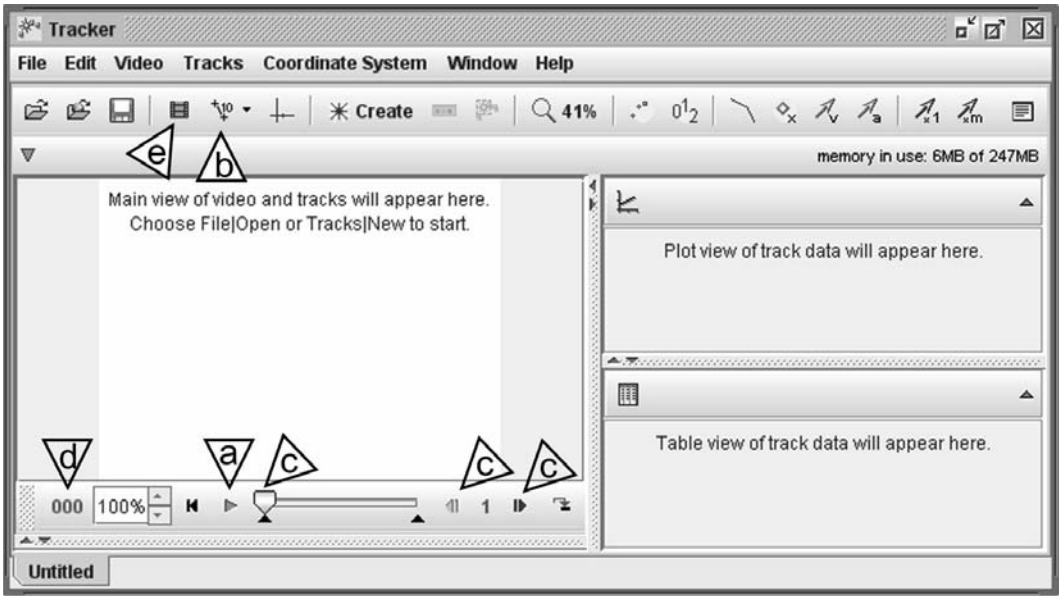


Fig. 6 The Tracker interface.

1. Download the videos onto the computer.

2. Run the Tracker program on the computer and open a video in it.

3. Play the video in the program, at **a** in **figure 6**. Repeat for the other videos and select the best video for subsequent analysis.

4. Create a new calibration tape (not stick) in the program - at **b** in **figure 6**. We will use the calibration tape to detemine the velocities of the discs. Calibrate the tape by dragging its endpoints to the ends of the image of the ruler in the window. Change the indicated length of the tape the real length of the ruler (click on the value next to it, type in the length and hit enter). Note that you can use the mouse wheel to zoom in or out of the picture. The picture can also be dragged in the window with the mouse after you have zoomed in on it.

Determining the velocities of the discs

The video consists of a lot of still pictures, called frames, taken in rapid succession at a constant rate. Pause the video and step through the frames with the controls in the window at **c** in **figure 6**. Note that the frame numbers are indicated at **d**. The frame rate can be viewed by opening the clip inspector at **e**. You can step forward and backwards through the video, but it seems that the graphics are not synced correctly initially when stepping backwards and then forwards again. It does synchronize after stepping a few frames forwards again. Just keep this in mind when you want to position on a particular frame in the video.

We are going to use the calibration tape to measure the small distances, Δs, the discs cover from one frame to the next frame. From these distances the speeds of the discs can be calculated via

$$v = \frac{\Delta s}{\Delta t}$$

where Δt is the time lapse from one frame to the next. This value stays constant throughout the video and can be obtained from the clip inspector window - **figure 6 e**. To specify the velocities of the discs all that needs to be added is the direction in which the discs were moving. This is measured by stretching the calibration tape along the path followed by the discs. The program indicates the direction of the tape at the top of the pane, the stretched length of the tape is also repeated there

Note that the program assumes that the discs traveled in the opposite direction if the endpoints of the tape are swopped around. The program gives the angles of the motion of the discs as negative if they moved downwards. These negative angles were measured clockwise with respect to the positive x-axis.

Now use the frames just before and just after the collission and determine the velocities of the discs just before, $\vec{v}_{S1}$, and just after, $\vec{v}_{S2}$ and $\vec{v}_{T2}$, the collision. Tabulate your values. Also make a quick drawing in your report book indicating the directions in which the discs were moving before and after the collision. Indicate the direction in which the positive x-axis is pointing.

The momenta of the discs

In this part of the experiment we represent the momenta with arrows, which you will draw in your book. Momentum is a vector quantity and we need special mathematics to handle vectors. Here we will treat them graphically. Other rigorous mathematical procedures exists to do calculations with vectors. Proceed by calculating the magnitudes (sizes) of the momenta of the discs before, $\vec{p}_{S1}$, and after the collision, $\vec{p}_{S2}$ and $\vec{p}_{T2}$ by measuring their mass with the digital scale. Tabulate the magnitudes together with their directions. The momentum vectors are now specified completely by these combined quantities.

Choose a suitable scale factor so the the length of the arrow representations of these momentum vectors will be not shorter than about seven centimeters. Calculate the length of the arrows and tabulate these together with previous values.

The momentum of the sytem of discs after the collision

Graphically add the momentum vectors of the discs after the collision, $\vec{p}_{S2}$ and $\vec{p}_{T2}$, in your book to obtain the momentum of the sytem of discs after the collision, $\vec{p}_2$.

Momentum conservation

We now have the resultant momentum of the system of discs after the collision, $\vec{p}_2$. If momentum was conserved this should be equal to the momentum of the system of discs before the collision.

Question 5 What momentum vector is the resultant momentum of the system of discs before the collision?

Lets rename this vector $\vec{p}_1$. Draw $\vec{p}_1$ and $\vec{p}_2$ next to each other each in their respective directions. Draw the arrows from a common point on the paper so that they form a V symbol.

Question 6 Does it seem that the momentum of the system of discs was conserved? Explain your answer.

Question 7 Give possible reasons if it was not conserved.

Now draw another arrow from the tip of $\vec{p}_1$ to the tip of $\vec{p}_2$ and label this vector $\Delta\vec{p}$. This vector represent the change in the momentum of the system. It was brought about by an external force, $\vec{F}_e$, acting on the system. We can write it symbolically as

$$\vec{p}_1 + \Delta\vec{p} = \vec{p}_2$$

From Newton's second law

$$\vec{F} = \frac{d\vec{p}}{dt}$$

we can now say that on the average that

$$\vec{F}_e = \frac{\Delta\vec{p}}{\Delta t_e}$$

That is the external force was acting on the system in the direction of $\Delta\vec{p}$ and we would have been able to evaluate it if we knew how long it acted, Δt_e - most likely the duration of the collision, which we unfortunately don't.

Conclusion

Physicists have so much confidence in the principle of conservation of momentum that they use it to postulate the existence of particles that went undetected in collisions arranged in particle accelerators. They know that such particles must exist in order to account for the "missing" momentum.

More information

See the webpages http://www.atlas.ch/etours.html for more information on the ATLAS experiment at the Large Hadron Collider (LHC) - http://home.web.cern.ch/topics/large-hadron-collider

Tracker

Tracker is a very powerful video analysis and modelling program that can be downloaded from

http://www.cabrillo.edu/~dbrown/tracker/ . There are also extensive help pages on the use of the program.

An instructional video on the use of Tracker is also available in the Practicals folder on your Physics clickUP web page.

The camera

The camera used in this experiment is the Canon Powershot A810 or A2300. Here you need to look up how to shoot videos in the Auto Mode. The Getting Started guide (the User guide gives more information on these topics) can be downloaded from

http://software.canon-europe.com/

and is also available on clickUP.

It might be a good idea to delete all pictures on the camera before you start with the experiment. This can be done by formatting the camera.

Safety precautions

Please be careful when you get up on the laboratory chair and again when you step off it. Rather support yourself with at least one hand on the bench while you get on and off. Place the leading foot in the middle of the chair and not off to the side.

HEA

Heat

Objective

To investigate the relationship between the amount of heat added to a certain amount of water and the change in its temperature.

Learning objectives

- Connecting a simple circuit containing a resistor and a switch.
- To measure the temperature of the water as it is heated by the resistor.
- To determine the amount of heat added to the water by the resistor.
- To establish the relationship between the temperature change of the water and the amount of heat added to it.
- To determine the specific heat of water.

Introduction

You are going to use a resistor to heat the water in this experiment. An ammeter will measure the current, I, in the resistor. The current will be kept constant during the heating process. The current is the amount of electric charge in coulombs that flows through the resistor per unit time (seconds). It follows that the total amount of charge that flows through the resistor in a time period t is

$$q(t) = It$$

Some of the kinetic energy of the electric current is transferred to the atoms in the resistor. This process is similar to that of a rope slipping through your hands while you are holding onto it. The effect is known as joule, ohmic or resistive heating. The amount of electrical energy, in joules, converted by the resistor per unit electric charge, in coulombs, flowing through it is the electric potential difference, V, in volt, over the resistor. It follows that the amount of electrical energy converted by the resistor in a time period t is

$$\begin{aligned} E &= q(t)V \\ &= ItV \end{aligned}$$

This converted electric energy is transferred as heat, Q_R, from the resistor to the water. We assume that all of the heat is transferred to the water or

heat lost by resistor = heat gained by the water

$$\therefore Q_R = Q_W$$

The heat gained by the water effects a change in its temperature, ΔT, according to

$$Q_w = m_w c_w \Delta T \textbf{ .... (1)}$$

m_w being the mass and c_w the specific heat of the water, which is about $4200\,J/kg \cdot K$.

We conclude that

$$\begin{aligned} Q_W &= Q_R \\ &= VIt \textbf{ .... (2)} \end{aligned}$$

Transposing **equation (1)** we get that

$$\Delta T = \frac{1}{m_w c_w} Q_W \text{ .... (3)}$$

We would thus expect a directly porportional relationship between the temperature, ΔT, change of the water and the amount of heat, Q_W, absorbed by it.

Apparatus

- A Polystyrene cup.
- A 1Ω, $5W$ resistor to heat the water, and two clothes pegs to fix it to the cup.
- A thermometer with $0{,}5\ ^\circ C$ divisions.
- A stopwatch for timing the heating process.
- An adjustable *15 V* DC supply that can deliver at least 5 *A*.
- A switch to switch the current in the circuit on or off.
- An electronic scale to determine the mass of the water.

Setup

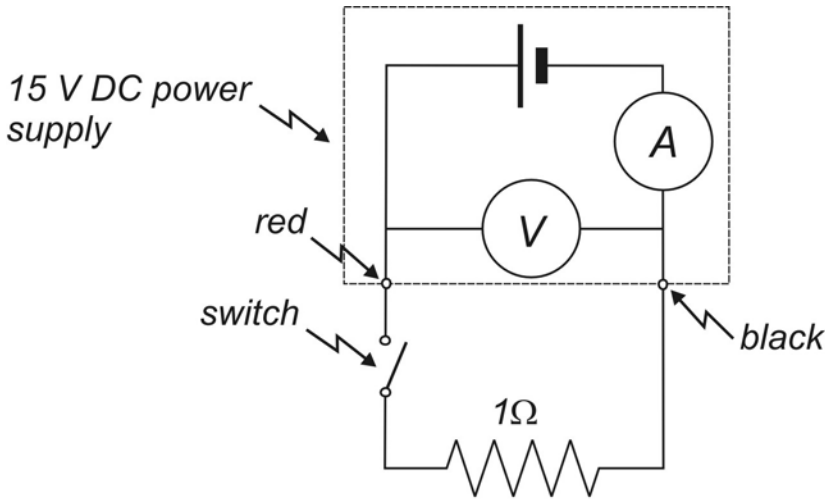


Fig. 1 The circuit for the heat experiment.

1. Check that the power supply is switched off and connect the 1Ω resistor and the switch in series to its black and red terminals with electrical leads (see **figure 1**).
2. Fill the cup to about half full with water and determine the mass, m_w, of the water.
3. Put the resistor into the water in the cup and fix it to the rim of the cup with clothes pegs.
4. Select the *5 A* setting on the power supply and switch it on. Turn the voltage down to zero.
5. Close the switch and adjust the voltage so that the current throught the resistor is between 2,5 *A* and 3,0 *A*.
6. Note the current, I, in and the potential difference, *V*, over the resistor in your report book.
7. Open the switch.

Measurements

Read through the following instructions before starting with the measurements. Make sure that you understand what you are supposed to do.

1. Measure the initial temperature, T_o, of the water after stirring it. Estimate fractional value of the temperature.

2. Make sure that all electrical connections are tight.

2. Close the switch by pushing the peg in deeply and start the stopwatch.

3. Keep on stirring the water while the resistor is heating the water. Record the temperature and time readings at one minute intervals for five minutes.

4. Open the switch.

Data analysis

1. Calculate and tabulate the temperature change, ΔT, of the water with respect to the initial temperature, T_o.

2. Also calculate and tabulate the amount of heat, Q_w, that was added by the resistor to the water for each of your measurements, using **equation (2)**. Use the current, I, and potential difference values, V, for the calculations.

3. Draw a graph that indicates the temperature change of the water, ΔT, as a result of adding an amount of heat, Q_w, to it.

4. According to your graph, what is the relationship between the variables ΔT and Q_w?

5. Write the relationship down with values and units for the constants in it. This is the experimentally determined relationship between ΔT and Q_w. We call it the empirical relationship.

6. Write **equation 3** down below your empirical relationship.

7. Compare the two formulas. What is the value of $1/m_w c_w$ according to your results?

8. Use this value to calculate the specific heat of water, c_w.

9. Calulate the percentage difference between your experimental value for c_w and the accepted value $4200\,J/kg \cdot K$.

10. What factors could have led to incorrect temperature values in the experiment?

Safety precautions

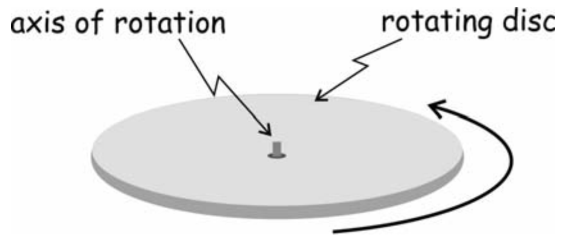
- Be careful not to break the thermometer. The resulting glass shards could cut you. Inform the demonstrator if you do break it.
- Be careful with the resistor. It gets quite hot and can burn you.

ROI

Rotational inertia

Objective

To experimentally determine the rotational inertia, I, or moment of inertia, of a flat disc rotating about a perpendicular axis through its center.

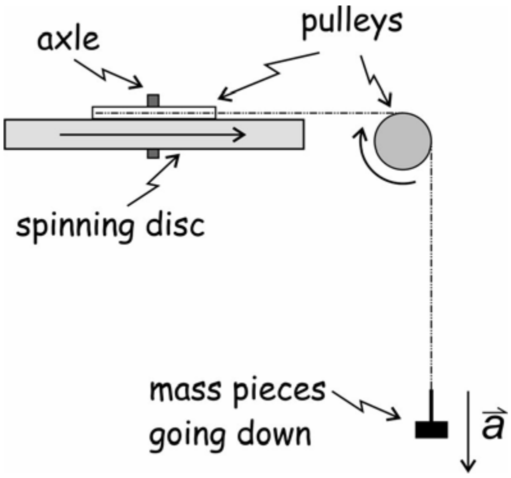


Learning objectives

- To apply a torque, τ_a (greek lower case tau symbol), to a disc and to determine its value.
- To measure the resulting angular acceleration, α, of the disc as produced by the torque.
- To draw an appropiate graph from which the moment of inertia, I, of the disc can be determined.
- To use the above mentioned graph to determine the opposing torque, τ_o, applied to the disc by friction.

Introduction

In this experiment you will use the weight, W, of a hanger and some mass pieces on it to apply a torque, τ_a, to the disc. The hanger is attached to a piece of thread that passes over a pulley and subsequently wrapped around another pulley mounted on top of the disc. The hanger accelerates downwards and creates a tension, T, in the thread which sets up the torque that spins the disc. You will vary the torque by altering the mass on the hanger and use a stopwatch to time the revolutions of the disc. From these measurements you can then calculate the applied torque and angular acceleration, α, of the disc.



Apparatus

- A base onto which the disc, with a pulley on top of it, is mounted horizontally with an axle - see **figure 1** on the next page.
- A spirit level for leveling the base and a short piece of wooden dowel.
- A vernier caliper for measuring the diameter of the pulley mounted on top of the disc.
- A ruler for measuring the diameter of the disc.
- Another separate pulley that can be mounted with a clamp on the edge of the laboratory work surface - the black pulley on the right-hand side of **figure 1**.
- A stopwatch for timing the disc's rotation.
- A mass hanger with some small masspieces and some thread.
- An electronic scale to determine the mass hanging from the thread.

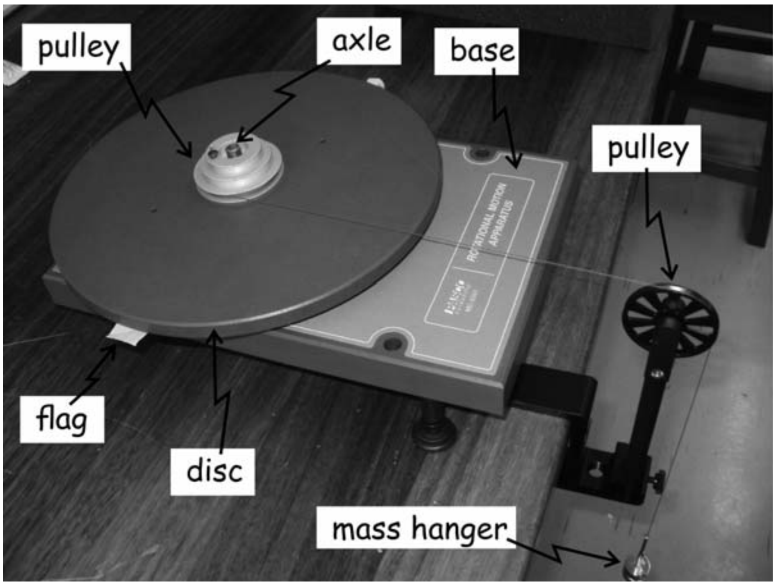


Fig. 1 The rotational inertia apparatus.

Theory

Before the practical consult Physics textbooks and internet resources in order to better understand the following

- How to measure the angular position, θ (in radians), of a rotating object
- Angular velocity, ω (in radians per second), of a rotating object
- Angular acceleration, α (in radians per second squared), of a rotating object
- Equations of motion for an object that is rotating with a constant angular acceleration
- What the rotational inertia or moment of inertia of a body measures
- Calculating the torque of a force acting on a body
- Newton's second law for rotation

The tension, $\vec{T}$, in the thread applies a torque, τ_a, to the disc according to

$$\tau_a = rT \quad \text{.....................} \ \textbf{(1)}$$

where *r* is the radius of the pulley mounted on top of the disc - **figure 2**. The same tension is applied to the hanging mass-pieces of mass *m*, as they move with an acceleration $\bar{a}$ downwards - **figure 3**. Applying Newton's second law to the accelerating mass we get

$$T = m(g - a) \quad \text{...............} \ \textbf{(2)}$$

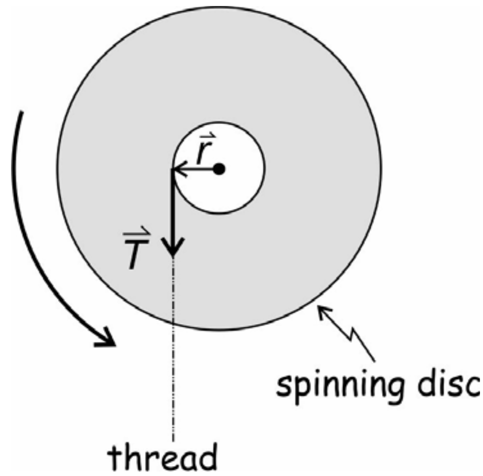


Fig. 2 The applied torque on the disc.

- see **figure 3**. The angular position, θ (in radians), of the disc is measured with respect to a line that remains fixed in a certain direction, line a in **figure 4**. The angular position of the disc is then the angle that a chosen line, rotating with the disc, line b in **figure 4**, makes with line a. The angular position of the disc continually increases as the disc spins around, since the angle between line b and line a increases with time. Note that the angular position does not revert back to zero when line b passes line a again.

If the angular acceleration of the disc, α, is constant, and the disc is accelerating from rest, and line b coincides with line a when it starts, then we can show from the rotational kinematics equations that the angular position, θ, of the disc after a time period, *t*, will be given by

$$\theta = \theta_o + \omega_o t + \tfrac{1}{2}\alpha t^2$$
$$\therefore \theta = \tfrac{1}{2}\alpha t^2 \quad \text{....................}\textbf{(3)}$$

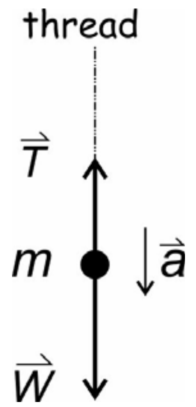


Fig. 3 Masspieces going down.

The relationship between the net torque, τ_{net}, on the disc and its angular acceleration, α, is given by Newton's second law for rotation

$$\tau_{net} = I\,\alpha$$

where I is the moment of inertia of the disc about the rotation axis. If friction applies a constant opposing torque, τ_o, to the disc, this becomes

$$\tau_a - \tau_o = I\,\alpha$$

where τ_a is the applied torque. It follows that

$$\tau_a = I\,\alpha + \tau_o \quad \text{....................}\ \textbf{(4)}$$

This is the linear relationship between the applied torque and the angular acceleration of the disc.

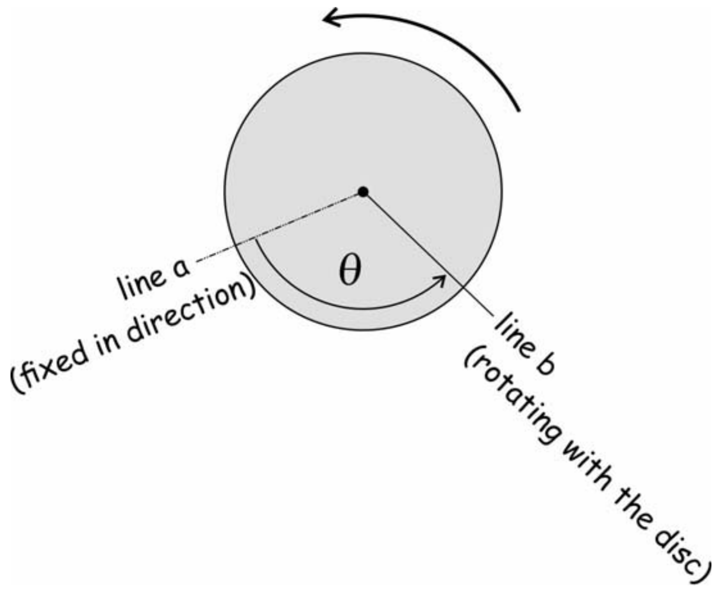


Fig. 4 The angular position of the rotating disc.

Setup

1. Clamp the black pulley with the mounting rod vertically onto the edge of the work surface.

2. Place the base just beyond the pulley on the worksurface and level it with the spirit level by screwing its adjustable feet into or out of the base - see **figure 1**.

3. Determine the mass of the mass hanger, m_h, the mass of the disc, m_d, and the diameter of the disc, D_d. Cut a piece of thread that is long enough to reach from the centre of the base to the floor when it passes over the black pulley. Tie one end to the mass hanger and the other end to the largest pulley on the disc .

4. Put the axle vertically in the bearing in the base and place the disc horizontally onto the axle.

5. Drape the thread over the black pulley and raise or lower the pulley in the clamp so that the thread runs parallel to the top surface of the disc. Change the angle of the black pulley so that it is parallel to the thread and not at an angle to it - see **figure 5**.

6. Keep the thread under tension while turning the disc by hand so that the thread winds onto the pulley on the disc. Stop when the hanger is just below the screw at the bottom of the clamp - see **figure 1**.

7. Hold the disc in this position and place the piece of dowel on the work surface just next to the flag on the disc .

8. Let go of the disc and count how many complete revolutions it makes before the hanger lands on the floor.

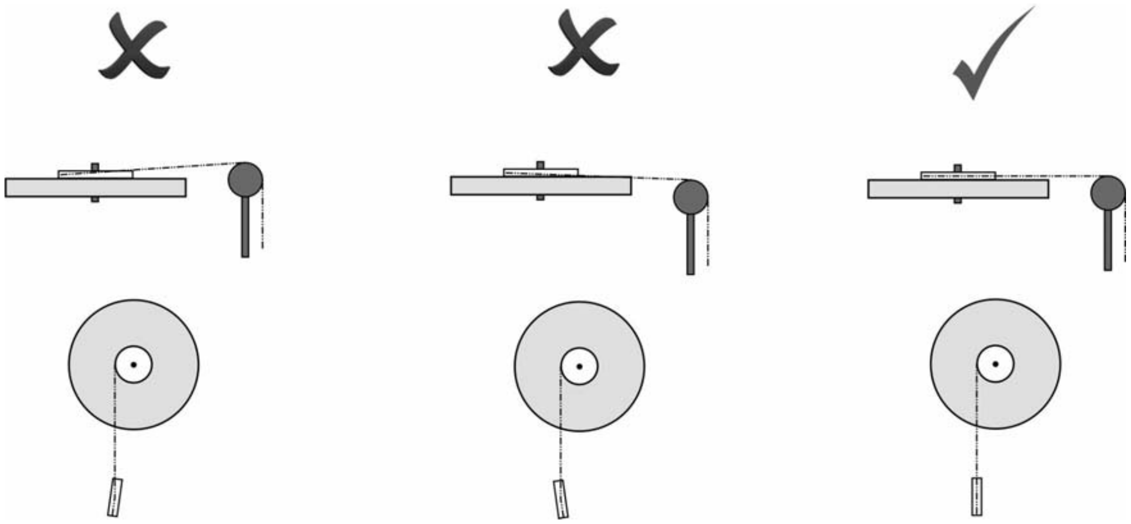


Fig. 5 Adjusting the height, top 3 figures, and direction, bottom 3 figures, of the pulley on the rod for optimal conditions.

Measurements

Use the stopwatch to measure the elapsed time for the disc to complete a few revolutions, N, out of rest. Stop the stopwatch before the hanger reaches the floor. Repeat the measurement three more times and calculate the average time, t_a. Keep the amount of revolutions the same for each measurement. Make another three such measurements, each time with more mass on the hanger. Note that the angular position of the disc, θ in radians, increases by 2π radians for each revolution it completes. Tabulate your measurements and values. Also measure the diameter of the pulley on the disc, D_p , with the vernier caliper.

Data analysis

Question 1 Explain how you can use the time measurements, t_a, to calculate the downward acceleration , a, of the mass pieces. Assume that the acceleration is constant.

Construct a table with columns for the accelerating mass, m, the average time, t_a, the downwards acceleration, a, the tension in the string, T, the applied torque to the disc, τ_a in newton meter, and the angular acceleration, α in radians per second square, of the disc. Supply units for each of these variables and also a caption below the table.

Calculate and write the values of the downwards acceleration, a, of each of the mass pieces in the table.

Question 2 Explain how you can calculate the tension, T, in the thread while the mass pieces are going down.

Calculate and tabulate the tension in the string, T, for each of the mass pieces as they are going down.

The tension in the string is creating a turning effect on the disc - see **figure 2**. We call this the torque, τ_a, with units newton·meter. Calculate the applied torque on the disc - **equation (1)** - for each of the tensions in the string and fill the values in on your table.

This turning effect, or torque, causes an angular acceleration, α, of the disc which we observes as the disc spinning faster and faster as the mass pieces are going down. We are going to use **equation (3)** to calculate it

$$\theta = \tfrac{1}{2}\alpha t^2 \quad \textbf{.....................(3)}$$

Here θ is the amount by which the angular position of the disc changed during the time period t. We are going to calculate the angular position of the disc from the start of its motion up to the measured time periods t_a. That means we need to calculate by how much the angular position of the disc advanced during N revolutions. Since the angular position of the disc advances by 2π radians for each revolution the angular position of the disc after a time period t_a will be $N2\pi$ radians. Calculate the angular acceleration of the disc for each of the measurment and fill the values into the table.

Now draw a graph of the applied toque on the disc, τ_a, against its angular acceleration, α.

Determine the gradient, *grad*, and y-intercept, y_o, of the graph.Supply values and units for these. Note that radians is actually a dimensionless unit since it is calculated by a distance/distance or more precisely

$$\theta = \frac{s}{r}$$

where s is the arc length and r the radius of a circle - see

http://www.mathsisfun.com/geometry/radians.html

for more information. That means that it is only necessary to actualy write the units *rad* to indicate that angles were measured in radians and not degrees, otherwise it can be omitted since it is a dimensionless quantity. Write down the equation of your graph in the form

$$\tau_a = grad\,\alpha + y_o$$

here τ_a and α are variables but *grad* and y_o are constants. Supply values and units (units in brackets) for these two constant in the equation of your graph. Expand the units of *grad* into the basic S.I units in order to see if some of them do cancel out. This is called the empirical equation. Compare the empirical equation to **equation (4)**. What seems to be the values and units of the moment of inertia I and the opposing torque, τ_o?

Conclusion

We used this experiment to measure the moment of inertia of the disc about a perpendicular axis through its center. For such a regularly shaped object we can derive a formula for its value

$$I = \tfrac{1}{2}MR^2$$

where M is the mass of the disc and R its radius. The same experimental procedure can be used to determine the moment of inertia of a more intricately shaped object.

Appendix A

The vernier caliper

The vernier caliper is an instrument that can make very accurate linear measurements. It was introduced in 1631 by Pierre Vernier of France. It utilises two graduated scales: a main scale similar to that found on a ruler and another graduated auxiliary scale, the vernier scale, that slides parallel to the main scale and enables readings to be made to a fraction of a division on the main scale.
The webpage

http://www.phy.uct.ac.za/courses/phylab1/current/vernier1.html

gives more information about the use of the vernier caliper.

The vernier caliper has two sets of jaws, one for measuring outer measurements (like the outer diameter of a disc) and one for inner measurements (like the inside diameter of a pipe). It also has a pin for measuring depths (like the depth of a groove) - see **figure 1**.

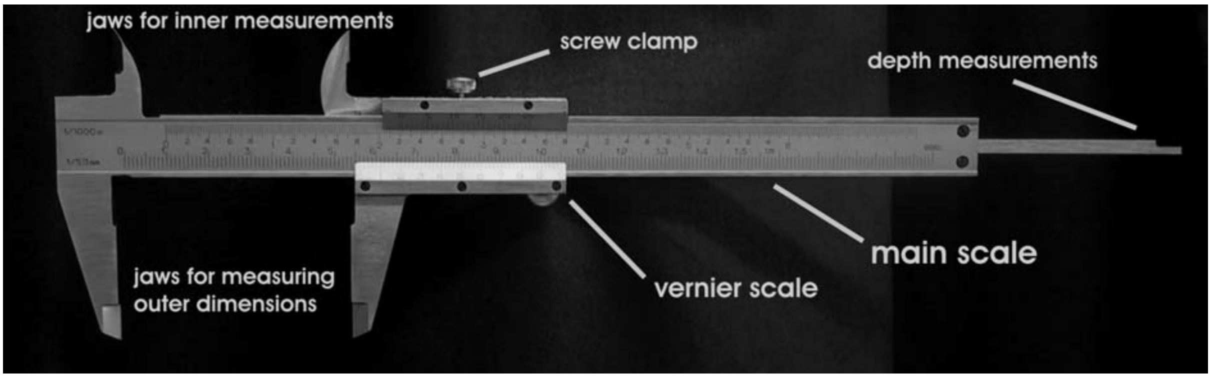


Fig. 1 a Vernier Caliper.

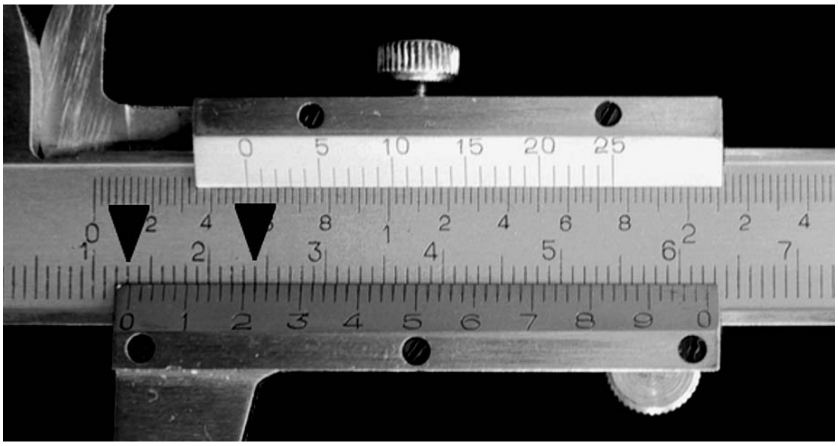


Fig. 2 Taking a reading on a vernier caliper.

The caliper in **figure 2** has 50 divisions on the vernier scale. The leftmost one, the index line, indicates the reading on the main scale, 13 mm in this case since the index line is just to the right of it. The reading on the vernier scale is taken at the position where a vernier line lines up with a line on the main scale. This is at the line just to the right of the 2 line on the vernier scale. This corresponds to a vernier reading of 0,22 mm. The caliper reading is therefore 13,22 mm (for all three types of measurements - outer, inner and depth measurements).

The meaning of the vernier divisions can be inferred by closing the caliper completely and noting that the 50 divisions on it are 49 mm long. One vernier division is therefore 0,98 mm wide. The first vernier division is therefore 0,02 mm short of the first millimeter mark on the main scale, while the second one is 0,04 mm short of the second millimeter mark on the main scale. If the jaws were therefore to be opened 0,02 mm the first vernier line will coincide with first millimeter line on the main scale, and if it were to be opened 0,04 mm the second vernier line will coincide with the second line on the main scale...

In **figure 3** the outer diameter of a cylinder is measured. The jaws were closed on it and the clamping screw tightened.

The smallest division on the vernier in **figure 3** is 0,02mm shorter than one millimeter.

The reading on the main scale ①is 7mm

The reading on the vernier scale ②is 0,28mm

The diameter of the cylinder is therefore 7,28mm

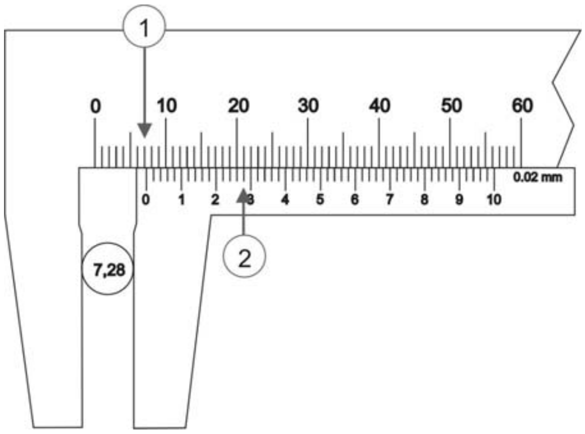


Fig. 3 Measuring with a vernier caliper

In order to determine the diameter it is therefore necessary to take readings on both the main- and vernier scales and combine these two readings. The vernier divisions of the caliper in **figure 4** are 0,05 mm shorter than two

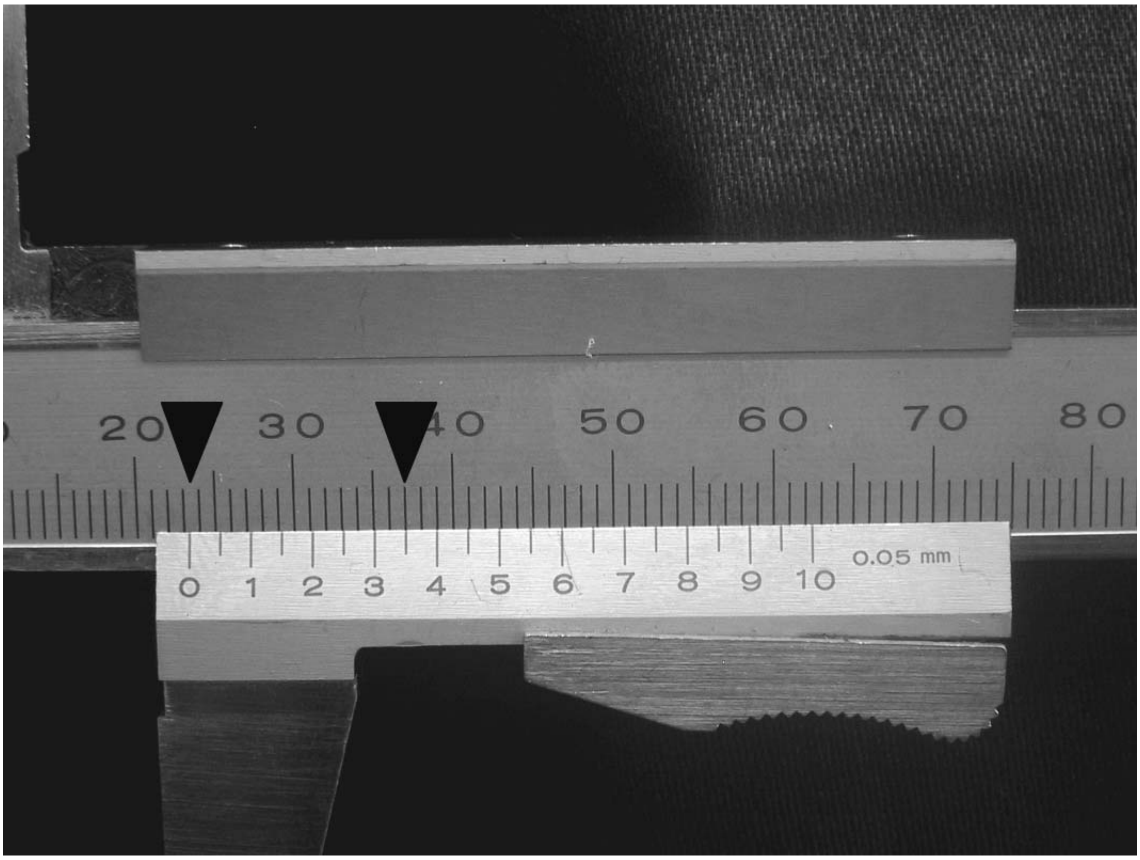
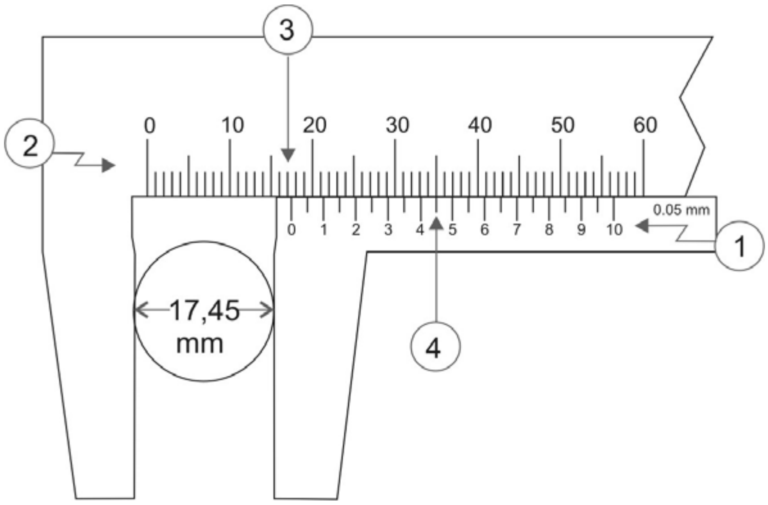


Fig. 4 Taking a reading on a vernier caliper.

millimeters. The reading on the main scale is 23 mm, while the vernier reading is 0,35 mm. The caliper reading is therefore 23,35 mm.

In **figure 5** another measurement is made with a similar caliper:



<u>Fig. 5</u> Reading of a vernier caliper

The vernier ①is the auxiliary scale that slides along the main scale ②.

The reading on the main scale is taken just to the left of the zero on the vernier scale ③(17mm in this case).

The reading on the vernier is taken where the marks on the vernier scale and the main scale coincide ④ (0,45mm in this case).

The final reading is the sum of readings on the vernier and the main scale.

The diameter of the object is therefore 17,45mm.

Appendix B

Electronic scale

METTLER TOLEDO PB1502

Fig. 1 METTLER electronic scale.

- Check that the security cable (**2** in **figure 1**) is not touching the scale pan and that the scale is not standing on the cable.
- Ensure that the scale is level (**1**). If not, screw the feet below the front of the scale in (anticlockwise) or out (clockwise).
- The scale is switched on and zeroed by pressing the **On** switch (**3**).
- Place the object (< 1,5 kg) on the scale pan and read the mass in grams.
- Note that the scale is sensitive to draughts due to the large scale pan (Bernoulli effect).

Appendix C

Stopwatch

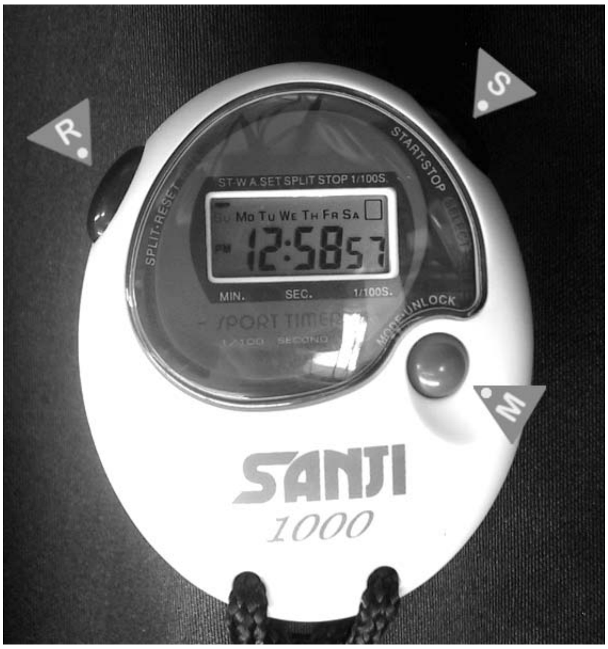


Fig. 1 A SANJI stopwatch.

There are several types of digital stopwatches in the Physics laboratory, but most of them are operated in the same way.

The first step is to get it in the stopwatch mode, since it can be in various modes like time (watch), altering the time and setting the alarm. Use the mode button (**M** button in the figures) to change it to the stopwatch mode. In the stopwatch mode you will see a line of segments at the top of the display flashing while the digits are not flashing. Once in this mode you can reset the stopwatch to zero with the Reset button (**R** in the figures). The stopwatch is started and stopped with the **S** button.

12:34 56

Fig .2 An example of a reading on a stopwatch.

Figure 2 shows an example of a reading on a stopwatch. The elapsed minutes appear to the left of the colon character (**:**) . To the right of the colon is the seconds. The smaller digits are the decimal seconds. The reading is therefore 12 minutes and 34,56 seconds.

Please note that it is a good practice to use the index finger rather than the thumb to press the start button when taking measurements, since it is generally the more agile one of the two.

It is possible to “freeze” the display while the stopwatch keeps on running in the background. This is achieved by pressing the **LAP** or **SPLIT** button (**R** in **figures 1** and **3**). The normal mode of the stopwatch is restored by pressing this button again.

Fig. 3 Another SANJI stopwatch.